

Astroparticle Physics

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These notes are not endorsed by the lecturers. All errors are mine.

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1. Introduction

Astroparticle physics is a very broad, yet niche subject. As the name suggests we want to use particle physics to explain astro-phenomena and the Universe. But we also want to use observations from astronomy to explain particle-phenomena. As an example we know from particle physics how very energetic particles behave and what light they can emit. When doing observations we see this light and can work backwards to determine or try to understand which physical processes lead to the emission. Which in turn tells us a great deal about the object we are observing.

Another part of this course is *multimessenger astronomy*, where we look at different *messengers* to say something about a given object. As an example protons do not travel freely in space since they are charged. This means they do not point back to their source. But, photons and neutrinos do. Using observations of photons and neutrinos might then tell us something about the origin of the proton.

2. Basics of Energy Loss

2.1. Ionisation

When high energy particles pass through matter they can lose energy by ionising or exciting atoms and molecules in said matter. This loss of energy influences the propagation of these particles in the Universe and leads to a heating of the interstellar medium. We can use this to detect particles and measure their properties.

We start by considering the non-relativistic case where a charged particle with mass M , charge Ze and velocity v scatters off an electron at rest with some impact parameter b and scattering angle θ . Assuming particles are hard spheres the maximal velocity of the electron after scattering is given by

$$v_{e,\max} = \frac{2M}{M + m_e} v \simeq 2v$$

The fractional energy loss of the particle is

$$\frac{\frac{1}{2}m_e(2v)^2}{\frac{1}{2}Mv^2} = \frac{4m_e}{M} \stackrel{M \gg m_e}{\ll} 1$$

This means the energy loss per collision is very small so we can assume the particles movement is unchanged due to the collision. The scattering is mediated by

$$|\mathbf{F}| = \frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r^2}$$

Consider the momentum impulse given to the electron

$$P \equiv \int_{-\infty}^{\infty} \mathbf{F} dt \stackrel{\text{by symmetry}}{=} \int_{-\infty}^{\infty} F_{\perp}$$

We use

$$F_{\perp} = \frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r^2} \sin \theta$$

and $v dt = dx$. With $x \tan \theta = b$ and $r \sin \theta = b$ we eventually find

$$p = \frac{Ze^2}{2\pi\epsilon_0 bv}$$

Then the kinetic energy transferred to the electron is

$$\frac{p^2}{2m_e} = \frac{Z^2 e^4}{8\pi^2 \epsilon_0^2 b^2 v^2 m_e}$$

We want the average energy loss per unit length our particle will experience moving through a material with electron density n_e . We consider a cylindrical geometry with the impact parameter being the radial coordinate b and the height being dx . The volume element of interest is then $dV = 2\pi b db dx$ and we find

$$\begin{aligned} \underbrace{-\frac{dE}{dx}}_{\text{energy loss}} &= n_e \int_{b_{\min}}^{b_{\max}} \frac{p^2}{2m_e} 2\pi b db \\ &= \frac{Z^2 e^4 n_e}{4\pi \epsilon_0^2 v^2 m_e} \int_{b_{\min}}^{b_{\max}} \frac{db}{b} \\ &= \frac{Z^2 e^4 n_e}{4\pi \epsilon_0^2 v^2 m_e} \ln\left(\frac{b_{\max}}{b_{\min}}\right) \end{aligned}$$

To estimate $b_{\max}$ we relate the collision time to the orbital period of the electron. As in this case the electron would only feel a slowly varying field and no ionisation takes place. Then

$$\tau \simeq \frac{2b_{\max}}{v} \simeq \frac{2\pi}{\omega_0}$$

implying

$$b_{\max} \simeq \frac{\pi v}{\omega_0}$$

For $b_{\min}$ we can use

$$\frac{p_{\max}^2}{2m_e} \simeq \frac{1}{2} m_e (2v)^2$$

However, this breaks for high velocities. To do better we use

$$\Delta p = mv_{\max} = 2vm$$

and

$$\Delta p \Delta x \gtrsim \hbar$$

Then we find

$$b_{\min} \sim \Delta x \geq \frac{\hbar}{2m_e v}$$

Using these guys we finally obtain

$$-\frac{dE}{dx} = \frac{Z^2 e^4 n_e}{4\pi \varepsilon_0^2 v^2 m_e} \ln \left(\frac{2\pi m_e v^2}{\hbar \omega_0} \right)$$

Assuming the atom is in the ground state then

$$I \equiv \frac{\hbar \omega_0}{2}$$

would be the ionisation energy. In reality we need the average $\bar{I}$ which is hard to compute.

Given this we write

$$-\frac{dE}{dx} = \frac{Z^2 e^4 n_e}{4\pi \varepsilon_0^2 v^2 m_e} \ln \left(\frac{m_e v^2}{\bar{I}} \right)$$

This is the non-relativistic result.

We denote the rest frame of M by S' and the rest frame of the electron by S . We assume $\mathbf{B} = 0$ so the only thing that happens is that $E_{\perp}$ is enhanced by the Lorentz factor γ . At the same time the collision time decreases by γ^{-1} . Then

$$\tau \sim \frac{2b}{v} \sim \frac{2b}{\gamma \beta c}$$

implying

$$b_{\max} \sim \frac{\pi \gamma \beta c}{\omega_0}$$

And

$$b_{\min} \sim \frac{\hbar}{\Delta p} = \frac{\hbar}{\gamma m v} = \frac{\hbar c}{\gamma m \beta}$$

This is the only thing that changes since everything else cancels. Then

$$-\frac{dE}{dx} \sim \ln \left(\frac{2\gamma^2 \beta^2 m_e c^2}{\bar{I}} \right)$$

A more correct treatment gives what is called the *Bethe formula*

$$\left\langle -\frac{dE}{dx} \right\rangle = K \underbrace{z^2}_{\text{charge}} \underbrace{\frac{\bar{Z}}{A}}_{\text{of absorber}} \frac{1}{\beta^2} \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 \gamma^2 W_{\max}}{\bar{I}^2} \right) - \underbrace{\beta^2 - \frac{\delta(\beta\gamma)}{2}}_{\text{density correction due to polarization}} \right]$$

with

$$K = 4\pi N_A r_e^2 m_e c^2$$

$$W_{\max} = \frac{2m_e c^2 \beta^2 \gamma^2}{1 + (2\gamma m_e)/M + (m_e/M)^2} \xrightarrow{2\gamma m_e \ll M} 2m_e c^2 \beta^2 \gamma^2$$

Typically this is shown against $\beta\gamma = p/Mc$.

2.2. Bremsstrahlung

Ionisation losses as described above underestimate the energy loss of relativistic electrons.

This additional energy loss is attributed to the acceleration of electrons in the electric field of an atom. This type of scattering amounts to flipping the picture described in the previous section. This acceleration leads to the emission of radiation called *bremsstrahlung* and the energy loss due to this radiation is given by *Larmor's formula*

$$-\frac{dE}{dt} = \frac{|\ddot{\mathbf{p}}|^2}{6\pi\epsilon_0 c^3} = \frac{q^2 |\mathbf{a}_0|^2}{6\pi\epsilon_0 c^3}$$

where $\mathbf{p} = q\mathbf{d}$ is the electric dipole moment and $\mathbf{a}_0$ is the proper acceleration. We need to rewrite this in the lab frame. Before seeing how this is done consider

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{dE}{dt} dt &= \frac{q^2}{6\pi\epsilon_0 c^3} \int_{-\infty}^{\infty} |\dot{\mathbf{v}}(t)|^2 dt \\ &\stackrel{\text{Parseval}}{=} \frac{q^2}{6\pi\epsilon_0 c^3} \int_{-\infty}^{\infty} |\dot{\mathbf{v}}(\omega)|^2 d\omega \end{aligned}$$

$$\begin{aligned}
&= \int_0^\infty \frac{q^2}{3\pi\epsilon_0 c^3} |\dot{\mathbf{v}}(\omega)|^2 d\omega \\
&= \int_0^\infty I(\omega) d\omega \\
&= \text{total emitted radiation}
\end{aligned}$$

Where we have defined the *emitted spectrum*

$$I(\omega) \equiv \frac{q^2}{3\pi\epsilon_0 c^3} |\dot{\mathbf{v}}(\omega)|^2$$

This is the quantity we would like to find. We proceed by considering the four-acceleration

$$\begin{aligned}
a_\mu &= \gamma \left(c\dot{\gamma}, \frac{\partial}{\partial t}(\gamma \mathbf{v}) \right) \\
&= \left(\gamma^4 c \left(\frac{\mathbf{v} \cdot \mathbf{a}}{c^2} \right), \gamma^2 \mathbf{a} + \gamma^4 \mathbf{v} \left(\frac{\mathbf{v} \cdot \mathbf{a}}{c^2} \right) \right)
\end{aligned}$$

with $\mathbf{a}$ and $\mathbf{v}$ measured in the lab frame S . In the rest frame S' we just have

$$a'_\mu = (0, \mathbf{a}_0)$$

We want to relate $\mathbf{a}_0$ with $\mathbf{a}$ and $\mathbf{v}$. Using $a_\mu a^\mu = a'^\mu a'_\mu$ we find

$$|\mathbf{a}_0|^2 = \gamma^4 \left(|\mathbf{a}|^2 + \gamma^2 \left(\frac{\mathbf{v} \cdot \mathbf{a}}{c} \right)^2 \right)$$

Then

$$-\frac{dE}{dt} = \frac{q^2 \gamma^4}{6\pi\epsilon_0 c^3} \left[|\mathbf{a}|^2 + \gamma^2 \left(\frac{\mathbf{v} \cdot \mathbf{a}}{c} \right)^2 \right]$$

where we have used

$$\frac{dE}{dt} = \underbrace{\frac{dE}{dE'}}_{\gamma} \underbrace{\frac{dE'}{dt'}}_{\gamma^{-1}} = \frac{dE'}{dt'} \sim \text{Lorentz invariant}$$

Writing $\mathbf{a} = \mathbf{a}_\parallel + \mathbf{a}_\perp$ we obtain

$$[\dots] = |\mathbf{a}_\parallel|^2 + |\mathbf{a}_\perp|^2 + \gamma^2 \frac{(v a_\parallel)^2}{c^2}$$

$$= |\mathbf{a}_\perp|^2 + \gamma^2 |\mathbf{a}_\parallel|^2$$

So we are left with

$$-\frac{dE}{dt} = \frac{q^2 \gamma^4}{6\pi \varepsilon_0 c^3} \left(|\mathbf{a}_\perp|^2 + \gamma^2 |\mathbf{a}_\parallel|^2 \right)$$

To determine the acceleration we use $\mathbf{F} = q\mathbf{E}$ to find

$$a_\parallel = \dot{v}_x = -\frac{ZeE_x}{m_e}$$

$$a_\perp = \dot{v}_z = -\frac{ZeE_z}{m_e}$$

Determining E_x and E_z is very annoying due to relativistic business. We eventually find

$$a_\parallel = \dot{v}_x = \frac{\gamma Ze^2 vt}{4\pi \varepsilon_0 m_e [b^2 + (\gamma vt)^2]^{3/2}}$$

$$a_\perp = \dot{v}_z = \frac{a_\parallel b}{vt}$$

We need the Fourier transform of these

$$\dot{v}_x = \frac{Ze^2}{4\pi \varepsilon_0 m_e} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\gamma vt}{[b^2 + (\gamma vt)^2]^{3/2}} e^{i\omega t} dt$$

Letting $b\xi \equiv \gamma vt$ we find

$$\dot{v}_x = \frac{Ze^2}{4\pi \varepsilon_0 m_e} \frac{1}{\sqrt{2\pi}} \frac{1}{bv\gamma} 2iy K_0(y)$$

where K_0 is the modified Bessel function and we have defined

$$y \equiv \frac{\omega b}{\gamma v}$$

Similarly we find

$$\dot{v}_z = \frac{Ze^2}{4\pi \varepsilon_0 m_e} \frac{1}{\sqrt{2\pi}} \frac{1}{bv} 2y K_1(y)$$

The emitted spectrum is then

$$I(\omega) = \frac{e^2}{3\pi \varepsilon_0 c^3} \left(|a_\parallel(\omega)|^2 + |a_\perp(\omega)|^2 \right)$$

$$\begin{aligned}
&= \frac{e^2}{3\pi\epsilon_0 c^3} \frac{1}{2\pi} \left(\frac{Ze^2}{4\pi\epsilon_0 m_e} \frac{1}{bv} \right)^2 \left(\frac{\omega b}{\gamma v} \right)^2 \\
&\times 2^2 \left[\frac{1}{\gamma^2} K_0^2 \left(\frac{\omega b}{\gamma v} \right) + K_1^2 \left(\frac{\omega b}{\gamma v} \right) \right]
\end{aligned}$$

For $y \ll 1$ or large v we have

$$K_0(y) \rightarrow \ln y$$

$$K_1(y) \rightarrow \frac{1}{y}$$

Meaning the $\perp$ contribution becomes constant and completely dominates the $\parallel$ contribution! We find

$$\begin{aligned}
I(\omega) &\stackrel{y \ll 1}{\simeq} \frac{2^2 e^2}{3\pi\epsilon_0 c^3} \frac{1}{2\pi} \left(\frac{Ze^2}{4\pi\epsilon_0 m_e} \frac{1}{bv} \right)^2 \\
&\simeq \frac{I_0}{2\pi} \frac{1}{b^2 v^2}
\end{aligned}$$

where we have defined

$$I_0 \equiv \frac{Z^2 e^6}{12\pi^3 \epsilon_0^3 c^3 m_e^2}$$

The last thing we need is to integrate over b . We have

$$\begin{aligned}
I(\omega') &= \int_{b'_{\min}}^{b'_{\max}} 2\pi b' \underbrace{\gamma n}_{\text{lab frame}} v \frac{I_0}{2\pi b'^2 v^2} db' \\
&= \frac{I_0 n \gamma}{v} \ln \left(\frac{b'_{\max}}{b'_{\min}} \right)
\end{aligned}$$

To estimate $b_{\min}$ we consider

$$\hbar \omega \stackrel{\text{maximal energy}}{=} \gamma m_e v^2$$

with $\tau \sim 2b_{\min}/v$ and $\omega' \simeq 2\pi/\tau$ we find

$$b_{\min} = \frac{\hbar}{m_e v}$$

To estimate $b_{\max}$ we consider when the charge of the nucleus gets screened by the electron cloud of the atom. By Thomas-Fermi the potential of the nucleus is

$$V(r) \simeq \frac{Ze^2}{4\pi\epsilon_0 r} e^{-r/a}$$

with

$$a \equiv \frac{1.4a_0}{z^{1/3}}$$

The screening distance is then $b_{\max} \sim a$. With everything we find

$$\begin{aligned} I(\omega) &= \frac{dE}{d\omega dt} \\ &= \frac{dE}{d\omega' dt} \frac{d\omega'}{d\omega} \\ &= I(\omega') \gamma^{-1} \\ &= \frac{I_0 n}{v} \ln \left(\frac{1.4a_0 m_e v}{z^{1/3} \hbar} \right) \end{aligned}$$

Then the total loss is

$$-\frac{dE}{dt} = \int_0^{E/\hbar} I(\omega) d\omega$$

where we integrate until the electron has given up all its energy in one collision. With $v \simeq c$ we find

$$-\frac{dE}{dt} = \frac{I_0 n}{\hbar c} E \underbrace{\ln \left(\frac{192}{z^{1/3}} \right)}_{\text{mainly screening}}$$

A more correct treatment by Bethe and Heitler gave

$$-\frac{dE}{dt} = \frac{z(z+1.3)e^6 n}{16\pi^3 \epsilon_0^3 m_e^2 c^4 \hbar} E \left[\ln \left(\frac{183}{z^{1/3}} \right) + \frac{1}{8} \right]$$

We see the energy loss is exponential. To quantify this we define the radiation length X_0 by

$$-\frac{dE}{dx} = -\frac{dE}{dt} \frac{1}{c} = \frac{E}{X_0}$$

The above is quite good for $z \gtrsim 4$ where

$$\frac{1}{X_0} \simeq 4 \left(\frac{\hbar}{m_e c} \right)^2 z(z + 1.3) \alpha^3 n \ln \left(\frac{183}{z^{1/3}} \right)$$

with α being the fine structure constant. We have found

$$\left\langle -\frac{dE}{dx} \right\rangle \simeq \frac{E}{X_0}$$

$$\stackrel{\text{total loss}}{=} \underbrace{a(E)}_{\text{ionisation}} + \underbrace{b(E)E}_{\substack{\text{bremsstrahlung} \\ + \text{others}}}$$

Sometimes the radiation length is written using $\xi_0 = \rho X_0$.

Consider the number of radiated photons

$$\frac{dN}{dt d\omega} = \frac{I(\omega)}{\hbar \omega}$$

since $I(\omega)$ is constant we get either few very high energy photons or many low energy photons.

2.3. Photon energy losses

We have

$$N = N_0 e^{-x/\lambda}$$

with λ being the mean free path length

$$\frac{1}{\lambda} = n\sigma$$

with σ being the cross-section which depends on the given process we are considering e.g. the photoelectric effect, Rayleigh scattering, Compton scattering, etc. The process we care about and which dominates at large energies (> 1 MeV) is pair-production

$$\gamma + \gamma \rightarrow e^+ + e^-$$

The cross-section for this process is

$$\sigma = \frac{7}{9} \frac{A}{N_A} \frac{1}{\rho X_0}$$

so

$$\lambda = \frac{9}{7} X_0$$

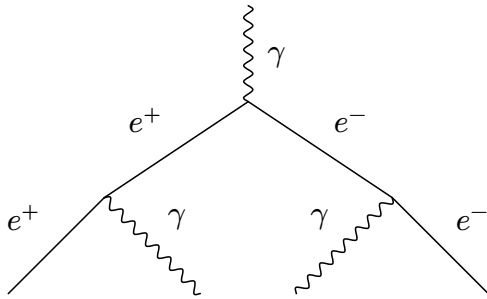
2.4. Electromagnetic showers

When high energy γ or $e^\pm$ enter an absorber they will initiate an *electromagnetic cascade*. The e^- will produce γ through bremsstrahlung which will then pair-produce $e^\pm$ again producing γ through bremsstrahlung. The γ just pair-produce $e^\pm$ directly. The difference in showers initiated by the two is the point of first interaction since $\lambda_\gamma > \lambda_e$.

As a simple model we assume the incoming particle has energy $E_0 \gg E_c$ with E_c being the *critical energy*. At energies above E_c the energy loss is dominated by bremsstrahlung and below it is dominated by ionisation. We assume $e^\pm$ travels X_0 before losing half its energy to a γ and γ travels X_0 before pair-producing $e^\pm$ with each carrying half the energy. We define

$$t = \frac{X}{X_0}; \quad \varepsilon = \frac{E}{E_c}$$

An example of a shower is:



After t radiation lengths we have $N(t) = 2^t$ and roughly equal amounts of γ and $e^\pm$.

The average energy is then

$$E(t) = \frac{E_0}{2^t}$$

The cascading stops when $E = E_c$ meaning

$$\frac{E_0}{2^{t_{\max}}} = E_c \Rightarrow t_{\max} = \frac{1}{\ln 2} \ln \frac{E_0}{E_c}$$

implying

$$N_{\max} = \frac{E_0}{E_c} \equiv y$$

A more sophisticated model gives

$$\frac{dE}{dt} = \frac{E_0 b (bt)^{a-1} e^{-bt}}{\Gamma(a)}$$

and

$$t_{\max} = \frac{a-1}{b} = \ln \frac{E_0}{E_c} - c_j$$

with $c_e = 1$ and $c_\gamma = \frac{1}{2}$. Another quantity sometimes used is the *shower age*

$$s = \frac{3t}{t + 2t_{\max}}$$

so $s_{\text{first}} = 0$, $s(t_{\max}) = 1$, and $s_{\text{death}} = 3$. The transversal development of the shower scales $\sim$ with the *Molière radius*

$$R_M \simeq \frac{21 \text{ MeV}}{E_c} X_0$$

Aside from these electromagnetic showers then *hadronic showers* are also possible. Here most particles end up being π with $\sim$ a third π^0 decaying into γ .

2.5. Cherenkov radiation

Cherenkov radiation is caused by the charged particles propagating in a medium with $\beta > n^{-1}$. When this happens a coherent waveform is created in similar fashion to a *Mach cone* for sound. We call this cone the *Cherenkov cone* and it is characterised by the *Cherenkov angle*

$$\cos \theta_c \equiv \frac{1}{n\beta}$$

The energy loss due to Cherenkov radiation is incredibly small and therefore negligible. However it is very important for particle detection as we will see later. The emitted spectrum for Cherenkov radiation follows

$$\frac{dN}{dx d\lambda} \simeq \frac{2\pi\alpha z^2}{\lambda^2} \sin \theta_c$$

importantly the emitted photons are optically visible.

3. Astrophysical Detectors

3.1. Cosmic rays

An example of a space-based cosmic ray detector is the *Alpha Magnetic Spectrometer* or *AMS-02*. The majority of onboard detectors abuse that cosmic rays are charged in various ways to determine composition and energy.

3.2. Gamma rays

An example of a space-based γ_{gamma} detector is the *Fermi Large Area Telescope* or *LAT* whose detection of γ_{gamma} is based on $\gamma \rightarrow e^{\pm}$. We like charged particles since we can easily track them. These therefore give us the direction of the original γ_{gamma} while a *calorimeter* later gives us the energy by abusing the electromagnetic cascades described above. The *Fermi Gamma-ray Burst Monitor* or *GBM* observes low energy γ_{gamma} by *scintillation detectors* which produce light when hit by a γ_{gamma} . The purpose of the GBM is to measure bursts of γ_{gamma} whereas the LAT measures and detects individual γ_{gamma} .

The majority of ground-based detectors are based on using the atmosphere as an inhomogeneous calorimeter whereing γ_{gamma} or cosmic rays (p, α, Fe , etc) can initiate showers. One such example is the *Pierre Auger* which measures both Cherenkov radiation with water detectors and the *greenish* light produced by deexcitations of nitrogen molecules in the air. *Imaging Cherenkov telescopes* in general try to measure the very brief light caused by Cherenkov radiation caused by some *primary particle*. This shows up as an ellipse (characterised by *Hillas parameters*) with the source being along the major axis of this ellipse. With multiple detectors one can therefore pinpoint the origin of the primary particle. An example of such a telescope is the *High Energy Stereoscopic System* or *H.E.S.S.*.

3.3. Neutrinos

The neutrino ν has very low mass and only interacts through gravity and weak interactions. This makes our job very difficult since they become annoying to detect. We care about ν since high-energy ν can be produced in hadronic processes initiated by cosmic rays. Typically through the decay of pions π^+ formed in proton-proton collisions or by cosmic rays interacting with γ . The nice thing about ν is that they are not charged meaning they act as *tracers* pointing back to their source. Then detecting them can tell us something about hadronic processes in said source. Alongside ν we always have the production of γ_{gamma} in hadronic processes. The Universe is also transparent to ν making them useful probes for the high-energy sky.

To detect ν we need a large detector since they interact weakly, and it should be built underground to isolate from cosmic rays and other background radiation. To make life simple we need a transparent medium such as ice or water at great depths. We want this since ν interacts with n and p in nuclei to form charged particles e.g. $\nu_\mu + N \rightarrow \mu + X$. These charged particles will propagate until they decay or interact. Assuming they are fast enough we can detect the Cherenkov radiation they emit. From the emitted spectrum many things can be inferred. This is the principle used by e.g. *IceCube* and *ANTARES*. The first observes two distinct types of *events* those being *tracks* or *cascades* with their meaning being relatively obvious. The important distinction is that cascade events tell us very little about where the ν came from while the track event does. With a cascade events we usually measure the γ being produced in the cascade whereas in track events we measure $\gamma_{\text{Cherenkov}}$.

3.4. Gravitational waves

Gravitational waves are simply *ripples* or small perturbations propagating through space-time at the speed of light. The amplitude of such a wave is characterised by the strain

$$h \equiv \frac{2 \Delta d}{d}$$

This is a measure of the relative change in distance between two objects due to the gravitational wave. These waves are formed by many things with the most famous example being *compact binary mergers* such as two black holes colliding.

We detect these waves by using laser interferometry. The most famous example of such a detector is the *Laser interferometer Gravitational-Wave Observatory* or *LIGO*. The idea is to pass a laser beam through a beam splitter creating two identical beams along two very long arms. At each end of an arm there is a mirror which reflects the beam ending with the beam merging at the beam splitter. When they merge the beams interfere before reaching a detector. These interferometers are calibrated such that these beams interfere destructively. Then any minute change in the length of the arms would lead to a change in interference and thereby a change in light intensity. The inteferometer is therefore sensitive to changes in the strain $h(t)$. The arms at LIGO are ~ 4 km long allowing it to detect incredibly small changes in length. Given these detectors are not directional, meaning they observe the entire sky at once we need multiple to deduce the direction of the source.

4. Non-thermal Emission Processes

4.1. Synchrotron radiation

We have seen that accelerating charged particles will emit bremsstrahlung as described by the Larmor formula. With *synchrotron radiation* the idea is similar. The emitted spectrum is typically low-energy and usually due to e^- .

We consider some particle in a uniform magnetic field. Then the equation of motion is¹

$$\frac{d}{dt}(\gamma m \mathbf{v}) = Ze \mathbf{v} \times \mathbf{B}$$

The LHS can be written as

$$\frac{d}{dt}(\gamma m \mathbf{v}) = \gamma m \mathbf{a} + \underbrace{m \mathbf{v} \frac{d\gamma}{dt}}_{\rightarrow 0}$$

where the second term vanishes since $\mathbf{a} \perp \mathbf{v}$ in a magnetic field. We write $\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp}$ with respect to $\mathbf{B}$ and define the *pitch angle* α by $|\mathbf{v}| = v_{\perp} \sin \alpha$ or

$$\tan \alpha = \frac{v_{\parallel}}{v_{\perp}}$$

Then $\mathbf{v}_{\parallel}$ does not change and we find ($a_{\parallel} = 0$)

$$\begin{aligned} \gamma m \mathbf{a} &= Ze v_{\perp} |\mathbf{B}| \hat{\mathbf{v}} \times \hat{\mathbf{B}} \\ &= Ze |\mathbf{v}| |\mathbf{B}| \sin \alpha \hat{\mathbf{v}} \times \hat{\mathbf{B}} \end{aligned}$$

implying $\mathbf{a} \perp \hat{\mathbf{v}}$ meaning we can equate

$$a_{\perp} = \frac{v_{\perp}^2}{r} = \frac{Ze |\mathbf{v}| |\mathbf{B}| \sin \alpha}{\gamma m}$$

we define the *gyro radius*

$$r_g \equiv \frac{\gamma m |\mathbf{v}| \sin \alpha}{Ze |\mathbf{B}|}$$

Then we find helical motion along $\mathbf{B}$! We define the *cyclotron frequency* by

¹Assuming $\mathbf{E} = 0$.

$$\omega_g = \frac{v_\perp}{r_g} = \frac{Ze|\mathbf{B}|}{\gamma m}$$

We can also write (using $p = |\mathbf{p}| = \gamma m |\mathbf{v}|$)

$$\begin{aligned} r_g &= \left(\frac{pc}{Ze} \right) \frac{\sin \alpha}{|\mathbf{B}|c} \\ &= \frac{R \sin \alpha}{|\mathbf{B}|c} \end{aligned}$$

where we define the *rigidity*

$$R \equiv \frac{pc}{Ze}$$

Consider a slowly varying magnetic field i.e. $\Delta B/B \sim \text{small}$ over $T = v_g^{-1}$. Then one can show

$$\frac{p_\perp^2}{B} = \text{const} \Rightarrow \Delta(Br_g^2) = 0$$

These are called *adiabatic invariants*.

We can determine the energy loss by

$$\begin{aligned} -\frac{dE}{dt} &= \frac{(Ze)^2 \gamma^4}{6\pi \varepsilon_0 c^3} \left(|a_\perp|^2 + \underbrace{\gamma |a_\parallel|^2}_0 \right) \\ &= \frac{e^2 \gamma^4}{6\pi \varepsilon_0 c^3} \frac{e^2 v^2 B^2 \sin^2 \alpha}{\gamma^2 m_e^2} \\ &= 2c \underbrace{\left(\frac{e^4}{6\pi \varepsilon_0^2 c^4 m_e^2} \right)}_{\sigma_T} \left(\frac{v}{c} \right)^2 \underbrace{\frac{B^2}{2\mu_0}}_{U_B} \gamma^2 \sin^2 \alpha \\ &= 2\sigma_T c U_B \beta^2 \gamma^2 \sin^2 \alpha \end{aligned}$$

We assume α is described by the distribution

$$P(\Omega_\alpha) d\alpha = \frac{2\pi \sin \alpha d\alpha}{4\pi} = \frac{1}{2} \sin \alpha d\alpha$$

with $0 \leq \alpha \leq \pi$. Then averaging we find

$$-\left\langle \frac{dE}{dt} \right\rangle = \frac{4}{3} \sigma_T c U_B \beta^2 \gamma^2$$

which is quite nice.

We would now like the emitted spectrum. Within the instantaneous rest frame S' of the particle then we will see the usual *dipole radiation*. We let the angle between $\mathbf{v}$ and the observer be ϕ' and the angle between $\mathbf{a}'$ and the observer θ' . We can then show that

$$I_\nu \propto \sin^2 \theta' = \cos^2 \phi'$$

which is the typical dipole radiation. By relativistic abberation we have

$$\begin{aligned} \sin \phi &= \frac{1}{\gamma} \frac{\sin \phi'}{1 + \beta \cos \phi'} \\ &\simeq \phi \\ &\stackrel{\text{highly relativistic}}{\simeq} \pm \frac{1}{\gamma} \end{aligned}$$

Meaning the radiation is strongly beamed forward in the lab frame S with opening ϕ .² We measure a radiation pulse whenever this elongated beam sweeps past the observer. Which happens when $\mathbf{v}$ is within an angle $\pm 1/\gamma$. The duration of this pulse can be found by considering two points A and B separated by $\theta \sim 1/\gamma$. The oberver receives emission when the particle is between these points. We have

$$t_A = \frac{R}{c}; \quad t_B = \frac{1}{c}(R - L) + \frac{L}{v}$$

implying

$$\Delta t = \frac{L}{v} \left(1 - \frac{v}{c} \right) \ll \frac{L}{v}$$

We can write

$$\frac{L}{v} = \frac{r_g \theta}{v} \simeq \frac{1}{\gamma \omega_g} \equiv \frac{1}{\omega_{g,\text{non-rel}}}$$

²We assume $\mathbf{v}$ is toward us. However this does not matter since if the particle was moving away from us the emitted radiation would be weakened.

and

$$1 - \beta \simeq \frac{1}{2\gamma^2}$$

implying

$$\Delta t \simeq \frac{1}{2\gamma^2\omega_{g,\text{non-rel}}}$$

The maximal Fourier component is then expected at the frequency

$$\nu \sim \frac{1}{\Delta t} \sim \gamma^2 \nu_{g,\text{non-rel}}$$

with

$$\nu_{g,\text{non-rel}} = \frac{eB}{2\pi m}$$

we call this the *critical frequency*.

Computation of the full spectrum is more complicated and requires using the *Liénard-Wiechert* potentials. We will not do this and simply state the result for a single e^-

$$I_{\perp}(\omega) = \frac{\sqrt{3}e^2\gamma \sin \alpha}{8\pi\epsilon_0 c}(F(x) + G(x))$$

$$I_{\parallel}(\omega) = \frac{\sqrt{3}e^2\gamma \sin \alpha}{8\pi\epsilon_0 c}(F(x) - G(x))$$

with

$$F(x) = x \int_x^{\infty} K_{5/3}(z) dz; \quad G(x) = x K_{2/3}(x)$$

where $x \equiv \omega/\omega_c = \nu/\nu_c$ and

$$\nu_c \equiv \frac{3}{2}\gamma^2 \nu_{g,\text{non-rel}} \sin \alpha$$

which is complicated. These represent the spectra radiated by a single e^- in the two polarisations during one period of the e^- orbit. The radiated power is then found by dividing with the orbital time

$$T_g \equiv \frac{1}{\nu_g} = \frac{2\pi\gamma m_e}{eB}$$

We find

$$P_{\perp}(\omega) = \frac{\sqrt{3}e^3 B \sin \alpha}{16\pi^2 \varepsilon_0 c m_e} (F(x) + G(x))$$

$$P_{\parallel}(\omega) = \frac{\sqrt{3}e^3 B \sin \alpha}{16\pi^2 \varepsilon_0 c m_e} (F(x) - G(x))$$

The total radiated power is then

$$P(\omega) = \frac{\sqrt{3}e^3 B \sin \alpha}{8\pi^2 \varepsilon_0 c m_e} F(x)$$

above $x = 1$ i.e. $\nu = \nu_c$ the spectrum has a sharp cutoff.

Consider an electron density that follows a power-law in energy

$$n \equiv \frac{dN}{dV d\gamma} = \kappa \gamma^{-p}$$

We want the *radiated emmissivity*

$$\begin{aligned} j(\nu) d\nu &= \frac{dE}{dV dt d\nu} d\nu \\ &= \left(-\frac{dE}{dt} \right) n(\gamma) \frac{d\gamma}{d\nu} d\nu \end{aligned}$$

with $\nu \sim \gamma^2 \nu_{g,\text{non-rel}}$

$$\begin{aligned} \frac{d\gamma}{d\nu} &= \frac{d}{d\nu} \left(\frac{\nu}{\nu_{g,\text{non-rel}}} \right)^{1/2} \\ &= \frac{\nu^{-1/2}}{2\sqrt{\nu_{g,\text{non-rel}}}} \end{aligned}$$

Then using $\nu_{g,\text{non-rel}} \sim B$ and $-dE/dt \sim \gamma^2 B^2$ we find

$$j(\nu) \sim \kappa \nu^{-\frac{p-1}{2}} B^{\frac{p+1}{2}}$$

Then the spectrum should follow a power-law with index

$$a = \frac{p-1}{2}$$

To be correct we would need to compute

$$j(\omega) = \int_1^\infty P(\omega, \gamma) n(\gamma) d\gamma$$

For $n(\gamma) \sim \gamma^{-p}$ this guy can be computed analytically. The total *luminosity* of some e^- filled volume can then be found by

$$L_\nu = \int_V j(\nu) dV$$

The total *flux* is then

$$F_\nu = \frac{L_\nu}{4\pi d_L^2}$$

We define the *fractional polarisation* by

$$\Pi(\omega) \equiv \frac{I_\perp(\omega) - I_\parallel(\omega)}{I_\perp(\omega) + I_\parallel(\omega)} = \frac{G(x)}{F(x)}$$

Then

$$\Pi = \frac{\int_0^\infty G(x) x^{\frac{p-3}{2}} dx}{\int_0^\infty F(x) x^{\frac{p-3}{2}} dx} = \frac{p+1}{p+\frac{7}{3}}$$

Typically we observe $p \sim 2.5$ giving $\Pi \sim 72\%$.

Synchrotron radiation comes alongside absorption due to γ interacting with charges in $\mathbf{B}$ giving up some energy. This is called *synchrotron self absorption*. We define the absorption coefficient κ_ν and it is given by

$$\kappa_\nu = \frac{p+2}{8\pi m_e \nu^2} \int_0^\infty d\gamma P(\nu) \frac{n(\gamma)}{\gamma} \sim B^{\frac{p+2}{2}} \nu^{-\frac{p+4}{2}}$$

The flux from a homogeneous ball of radiating plasma with radius r_b taking into account absorption is then

$$F_\nu = \frac{L_\nu}{4\pi d_L^2} \frac{3u(2r_b \kappa_\nu)}{2r_b \kappa_\nu}$$

Assuming $2r_b\kappa_\nu \ll 1$ i.e. optically thin we recover the previous while for $2r_b\kappa_\nu \gg 1$ i.e. optically thick we have

$$F_\nu \sim \frac{\nu^{-\frac{p-1}{2}}}{\nu^{-\frac{p+4}{2}}} = \nu^{5/2}$$

4.2. Inverse Compton scattering

We have *inverse Compton scattering* whenever relativistic e^- propagate through a γ gas. Whenever the e^- loses energy due to this scattering we have inverse Compton scattering.

Before continuing recall *Doppler boosting* which refers to the change in observed frequency of a moving source. We have

$$\nu = \delta_D \nu'; \quad \delta_D = \gamma^{-1}(1 - \beta \cos \theta)^{-1}$$

where δ_D is the *Doppler factor*. We now consider an e^- moving with β through an isotropic γ field in the lab frame S . Within the rest frame of the e^- denoted by S' all γ are incident within a narrow cone. A γ incident with θ in S will be incident with θ' determined by

$$\tan \theta' \simeq -\frac{1}{\gamma} \cot \frac{\theta}{2}$$

The γ energy in S' is

$$\varepsilon' = \frac{\varepsilon}{\delta_D}$$

implying

$$\varepsilon_{\min'} \sim \frac{\varepsilon}{2\gamma}; \quad \varepsilon'_{\max} \sim 2\gamma\varepsilon$$

Two limiting cases are important:

1. $\varepsilon' \ll m_e c^2 \sim$ *Thomson regime* where the e^- loses small amounts of energy continuously.
2. $\varepsilon' \gg m_e c^2 \sim$ *Klein-Nishina regime* where the e^- loses a large amount of energy in one scattering event.

After scattering we can compute the γ energy as

$$\epsilon'_1 = \frac{\epsilon'}{1 + (\epsilon'/m_e c^2)(1 - \cos \theta'_1)} \Rightarrow \epsilon_1 = \gamma \epsilon'_1 (1 + \beta \cos(\pi - \theta'_1))$$

Within the Thomson regime $\epsilon'_1 \simeq \epsilon'$ we have

$$\epsilon_{1,\max} \sim 2\gamma \epsilon'_{1,\max} \sim 4\gamma^2 \epsilon$$

This process is very efficient! The energy loss for an e^- accelerated by some incident radiation in the Thomson regime can be shown to be

$$-\frac{dE}{dt} = \sigma_T c u_{\text{rad}}$$

where u_{rad} is the *total radiation density*.

Then in S' we have

$$-\frac{dE}{dt} = -\frac{dE'}{dt'} = \sigma_T c u'_{\text{rad}}$$

where u_{rad} is isotropic in S . To transform u_{rad} we note that the total number of γ in some volume is invariant

$$dN(\epsilon) = n(\epsilon) dx dy dz \sim \text{invariant}$$

The spacetime volume is also invariant

$$dV_4 = dx dy dz dt \sim \text{invariant}$$

which can be seen explicitly by computing³

$$dt dx = \begin{vmatrix} \frac{\partial t}{\partial t'} & \frac{\partial x}{\partial t'} \\ \frac{\partial t}{\partial x'} & \frac{\partial x}{\partial x'} \end{vmatrix} dx' dt'$$

Then

$$\frac{dN(\epsilon)}{dV_4} = \frac{n(\epsilon)}{dt} \sim \text{invariant}$$

Since dt transforms the same as ϵ by

$$\epsilon = \delta_D \epsilon'; \quad t = \delta_D t'$$

Then we obtain

³Follows immediately since $\det g_{\mu\nu} = 1$ in Minkowski space.

$$\frac{n(\varepsilon)}{\varepsilon} \sim \text{invariant}$$

Consider now u'_{rad}

$$\begin{aligned} u'_{\text{rad}} &= \varepsilon' n' = \delta_D^{-2} \varepsilon n \\ &= \delta_D^{-2} u_{\text{rad}} = \gamma^2 (1 - \beta \cos \theta)^2 u_{\text{rad}} \end{aligned}$$

By assumption u_{rad} is isotropic so

$$\frac{du_{\text{rad}}}{d\Omega} = \frac{u_{\text{rad}}}{4\pi}$$

implying

$$\begin{aligned} \frac{du'_{\text{rad}}}{d\Omega} &= \delta_D^{-2} \frac{du_{\text{rad}}}{d\Omega} \\ &= \frac{\delta_D^{-2}}{4\pi} u_{\text{rad}} \end{aligned}$$

We can integrate to find

$$\begin{aligned} u'_{\text{rad}} &= \frac{u_{\text{rad}}}{4\pi} \gamma^2 \int d\Omega (1 - \beta \cos \theta)^2 \\ &= \frac{u_{\text{rad}}}{4\pi} \gamma^2 \int (1 - \beta \cos \theta)^2 \sin \theta d\theta d\phi \\ &= \frac{4}{3} u_{\text{rad}} \left(\gamma^2 - \frac{1}{4} \right) \end{aligned}$$

The energy lost by the e^- is the same as the energy gained by the γ giving

$$\frac{dE}{dt} = \frac{4}{3} \sigma_T c u_{\text{rad}} \left(\gamma^2 - \frac{1}{4} \right)$$

We also need to subtract the initial energy of the γ being $\sigma_T c u_{\text{rad}}$

$$\begin{aligned} \frac{d\varepsilon_1}{dt} &= \frac{4}{3} \sigma_T c u_{\text{rad}} \left(\gamma^2 - \frac{1}{4} - \frac{3}{4} \right) \\ &= \frac{4}{3} \sigma_T c u_{\text{rad}} \beta^2 \gamma^2 \\ &= -\frac{dE}{dt} \end{aligned}$$

This is the energy gained by γ in the Thomson regime. We notice this has the same structure as synchrotron radiation with the replacement $U_B \rightarrow u_{\text{rad}}$. This is the case because synchrotron radiation can be treated as scattering with virtual γ from $\mathbf{B}$. We can also find the average γ energy after scattering by

$$\begin{aligned} -\frac{dE}{dt} &= \langle \varepsilon_1 \rangle \frac{dn}{dt} \\ &= \frac{4}{3} \gamma^2 \beta^2 \varepsilon \underbrace{\sigma_T c \frac{u_{\text{rad}}}{\varepsilon}}_{\text{number of scattered } \gamma} \end{aligned}$$

implying

$$\langle \varepsilon_1 \rangle = \frac{4}{3} \gamma^2 \beta^2 \varepsilon$$

Due to the similarity with synchrotron radiation we expect a power-law behaviour

$$j(\nu) \sim \nu^{-\frac{p-1}{2}}$$

Additional notes:

1. The scattered radiation is polarised with

$$\Pi = \frac{1 - \cos^2 \theta}{1 + \cos^2 \theta}$$

2. Thomson scattering is very important for opacity. Assuming some medium has e^- density n_e then the γ density n_γ decreases as

$$-\frac{dn_\gamma}{dt} = \sigma_T c n_e n_\gamma$$

$$-\frac{dn_\gamma}{dx} = \sigma_T n_e n_\gamma$$

implying

$$n_\gamma = n_0 \exp \left[- \underbrace{\int \sigma_T n_e dx}_{\text{optical depth } \tau} \right]$$

The γ scatter and perform random walks with

$$\lambda = \frac{1}{\sigma_T n_e}$$

This is the cause of the *cosmic microwave background* where the opacity τ drops below 1.

Going from a single e^- to an e^- cloud is complicated. We consider a γ gas with γ density $dn/d\varepsilon$ and an e^- distribution with e^- density $dN_e/dV d\gamma$. We eventually find for e^- with energy γmc^2 and γ with energy ε

$$\frac{dN}{dt d\varepsilon_1 d\gamma d\varepsilon} = \frac{3\sigma_T c}{4} \frac{1}{\gamma^2} \left(\frac{1}{\varepsilon} \frac{dn}{d\varepsilon} \right) f\left(q = \frac{\varepsilon_1}{\Gamma_\varepsilon(1 - \varepsilon_1)}; \Gamma_\varepsilon = \frac{4\varepsilon\gamma}{m_e c^2}\right)$$

with

$$f(\varepsilon_1, \varepsilon, \gamma) = 2q \ln q + (1 + 2q)(1 - q) + \frac{1}{2} \frac{(\Gamma_\varepsilon q)^2}{1 + \Gamma_\varepsilon q} (1 - q)$$

Then the total *IC emissivity* is given by

$$j_{\nu_1}^{\text{IC}} = h\nu_1 \int d\gamma \int d\varepsilon \frac{dN}{dt d\varepsilon_1 d\gamma d\varepsilon} \frac{dN_e}{dV d\gamma}$$

To determine the energy loss in the Klein-Nishina regime we consider $\Gamma_e \gg 1$

$$\begin{aligned} -\dot{\gamma}^{\text{IC}} &= \frac{1}{m_e c^2} \int d\varepsilon_1 \int d\varepsilon (\varepsilon_1 - \varepsilon) \frac{dN}{dt d\varepsilon_1 d\gamma d\varepsilon} \\ &\simeq \frac{3}{8} \frac{\sigma_T}{m_e c^3} \int d\varepsilon \left(\frac{1}{\varepsilon} \frac{dn}{d\varepsilon} \right) \left(\ln \frac{4\varepsilon\gamma}{m_e c^2} - \frac{11}{6} \right) \end{aligned}$$

A black body γ gas with temperature T gives

$$-\dot{\gamma}^{\text{IC, KN}} \sim T^2 \ln \gamma T$$

While in the Thomson regime we find

$$-\dot{\gamma}^{\text{IC, T}} = \frac{4}{3} \frac{\sigma_T}{m_e c} \beta^2 \gamma^2 U_{\text{ph}} \sim \gamma^2 T^4$$

We see the cooling is slower in the Klein-Nishina limit!

We can assume

$$\frac{dN_e}{d\gamma} \sim \gamma^{-p}$$

as before. Then we find

$$j_{\nu}^{\text{IC, T}} \sim \nu^{-\frac{p-1}{2}}$$

$$j_{\nu}^{\text{IC, KN}} \sim \nu^{-p}$$

The simplest *synchrotron self Compton* energy distribution clearly exhibits these two limits and the synchrotron spectrum with self absorption. Here the e^- responsible for synchrotron radiation do inverse Compton scattering with the produced γ .

4.3. Hadronic processes

Aside from the above *leptonic* emission models which produce high energy γ_{gamma} rays we need *hadronic* models to explain observed ν . We consider binary collisions $pp \rightarrow \pi X$ and *photohadronic* interactions $N\gamma \rightarrow \dots$

We start with $pp \rightarrow \pi X$. Consider the distribution of produced *secondaries*⁴ per time per kinetic energy T_s

$$\dot{n}_s = \frac{dN_s}{dt dV dT_s}$$

Assuming we have an isotropic p distribution interacting with stationary target p with density n_p we have

$$\dot{n}_s(T_s) = 4\pi n_p \int_0^\infty dT_p J_p(T_p, \Omega_p) \frac{d\sigma_{pp \rightarrow \pi}}{dT_\pi}$$

where J_p is the cosmic-ray number intensity and $d\sigma_{pp \rightarrow \pi}/dT_\pi$ is the inclusive cross-section for π production.⁵ This depends on n_p which is typically small.

We have multiple photohadronic processes of interest:

$$N\gamma \rightarrow N\pi \sim \text{photo-pion production}$$

$$N\gamma \rightarrow Ne^+e^- \sim \text{photo-pair production}$$

$$N\gamma \rightarrow N'N'' \sim \text{disintegration}$$

⁴The π 's in this case.

⁵This is independent of X .

The cross-section depends on

$$\text{invariant} = E_r = \gamma_p E_\gamma (1 - \beta_p \mu) = E'_\gamma$$

E_γ is the γ energy in the p rest frame. We also have $\gamma_p = E_p/m_p c^2$ and $\mu = \cos \theta$. We consider a general γ field with

$$n_\gamma = \frac{dN_\gamma}{dE dV d\Omega}$$

Then the interaction rate is

$$\dot{N}_{\text{sc}} = c \oint d\Omega \int_0^\infty dE_\gamma (1 - \beta_p \mu) n_\gamma(E_\gamma, \Omega) \sigma_{\gamma p}(E_r)$$

The p loses a fraction $K(E_r)$ of its energy on average in each interaction we refer to this quantity as the *inelasticity*. Then

$$t_{\gamma p}^{-1} \equiv \left| \frac{\dot{\gamma}_p}{\gamma_p} \right| = c \int_0^\infty dE_\gamma \underbrace{\int_0^{2\pi} d\phi \int_{-1}^1 d\mu (1 - \beta_p \mu) n_\gamma(E_\gamma, \Omega) \sigma_{\gamma p}(E_r) K(E_r)}_{\oint d\Omega}$$

We generally have $t_{\text{sc}} = \dot{N}_{\text{sc}}^{-1} \ll t_{\gamma p}$. We have $\sigma_{p\gamma} \sim 10^{-2} \sigma_{pp}$ however typically $n_\gamma \gg n_p$ so

$$\lambda_{p\gamma} \ll \lambda_{pp}$$

implying $p\gamma \rightarrow \dots$ dominates.

The π 's produced will decay after being produced through the following *secondary production channels*

$$\begin{aligned} \pi^+ &\rightarrow \underbrace{\mu^+}_{e^+ + \bar{\nu}_\mu} + \nu_\mu \\ \pi^0 &\rightarrow \gamma + \gamma \end{aligned}$$

This leads to the observed ν !

We would like the spectrum of π^0 decay. Within the rest frame of the π^0 we have

$$E'_\gamma = \frac{m_\pi}{2} \simeq 67.5 \text{ MeV}$$

Given π^0 has spin zero the emitted γ have no preferred direction meaning their spectrum will be isotropic

$$\frac{dN}{d\Omega'} = \frac{N}{4\pi}$$

We transform to the rest frame of π^0 using β_π and $\gamma_\pi = E_\pi/m_\pi$.⁶ Then

$$\begin{aligned} E_\gamma &= \gamma_\pi E'_\gamma - \beta_\pi \gamma_\pi p'_\gamma \mu \\ &= \frac{m_\pi \gamma_\pi}{2} (1 - \beta_\pi \mu) \end{aligned}$$

implying

$$E_\gamma^{\min} = \frac{E_\pi}{2} (1 - \beta_\pi); \quad E_\gamma^{\max} = \frac{E_\pi}{2} (1 + \beta_\pi)$$

We have

$$\frac{dN}{dE'_\gamma d\Omega'} = 2 \frac{\delta(E'_\gamma - \frac{m_\pi}{2})}{4\pi}$$

and

$$\text{invariant} \sim \frac{1}{E_\gamma} \frac{dN}{dE_\gamma d\Omega}$$

Then $dN/dE_\gamma d\Omega$ transforms as $E'_\gamma = \delta_D^{-1} E_\gamma$ meaning

$$\frac{dN}{dE_\gamma d\Omega} = \frac{\delta(E'_\gamma - \frac{m_\pi}{2})}{2\pi \gamma_\pi (1 - \beta_\pi \mu)}$$

We integrate over $d\Omega = -d\phi d\mu$ with $\mu \rightarrow E_\gamma$ giving

$$\frac{dN}{dE_\gamma} = \frac{2}{\beta_\pi \gamma_\pi m_\pi} \Theta\left(E_\gamma, \frac{E_\pi}{2}(1 - \beta_\pi), \frac{E_\pi}{2}(1 + \beta_\pi)\right)$$

The final γ spectrum is found numerically.

⁶Using $c = 1$.

5. Extragalactic Sources

5.1. AGNs

Active galactic nuclei or *AGN* are observed to be extremely bright sources on the sky, typically across the entire electromagnetic spectrum. Historically the different types of sources say *radio-quiet* or *radio-loud* AGN were thought to be physically different objects. However, now it is generally believed that all the different types of AGN we observe stem from the same physical object. This viewpoint is referred to as *unification*. The unified model⁷ proposes all AGN are driven by *accretion* onto a central *supermassive black hole*.⁸ The black hole is surrounded by a hot accretion disk which produces thermal radiation in the optical and UV. Some AGN also produce large *jets* of relativistic particles carrying mass, energy, momentum, and magnetic fields away from the AGN.⁹ The accretion disk also acts to ionise the gas surrounding the black hole. This leads to the creation of two emission regions: the *broad line region* and the *narrow line region*. The main difference being their velocities leading to different amounts of Doppler shift. Lastly an *obscuring torus* composed of dust surrounds the central region. This torus is why various AGN can appear different even though they are truly the same object. The torus also absorbs UV radiation from the accretion disk and re-emits it in the infrared.¹⁰

Classification of AGNs is quite boring and needlessly complicated. We simply mention that *blazars* are AGN whose jets point toward us. This leads to very bright emissions across the entire electromagnetic spectrum, and therefore these are nice astrophysical laboratories. Some blazars have featureless, boring spectra, while others show strong,

⁷These unification models can be *strong* or *weak* depending on how many fundamental parameters they include. The strongest models say only *orientation* and *obscuration* determine the observed differences in AGN.

⁸This typically leads to *high variability* since black holes are relatively small.

⁹Why they do this is still a mystery. An explanation could be the *Blandford-Znajek mechanism* where the central black hole spinning is believed to produce an *EMF*.

¹⁰Typically referred to as the *infrared bump*.

broad emission lines. The boring ones are called *BL Lacs* and the exciting ones are called *FSRQs*.

5.2. Black holes

For our purposes a black hole is a region of space where a large amount of mass is concentration in a very small volume. This leads to a very strong gravitational field which even light cannot escape. The boundary surrounding the black hole is referred to as the *event horizon*.¹¹ These objects primarily form when massive stars collapse. However, very large black holes are believed to form in other ways. We are interested in black holes since these objects have very large gravitational fields on small scales. We might therefore get clues as to what a theory of quantum gravity should look like by studying black holes.

Hawking showed quantum effects near the horizon cause black holes to emit radiation, which we call *Hawking radiation*. This implies black holes have a temperature¹² and will evaporate. This is a problem since radiation is *thermal* by definition, meaning when the black hole evaporates we evolve from a *pure* state $|\Psi\rangle_{\text{BH}}$ to a *mixed* state $|\Psi\rangle_{\text{radiation}}$. This violates *unitarity* and is therefore a problem. We typically call this the *information paradox* since information is lost.

5.3. Blazars

As mentioned blazars are AGN whose jets point toward us. These AGN are radio-loud, show high variability, strong optical polarisation, and the familiar SED with a high-energy and low-energy peak. The low-energy peak is believed to be caused by synchrotron emission¹³ The high-energy peak is believed to be caused by inverse Compton emission. However, the latter is somewhat disputed. Some believe it is *leptonic* with e^- scattering

¹¹When something crosses the horizon it is destined to reach the center. However, for an outside observer nothing can ever cross the horizon.

¹²The *Hawking temperature*.

¹³The position of the peak further classifies blazars as *low synchrotron peak* or *LSP* blazars, *ISP* blazars, and *HSP* blazars.

with γ_{synch} and γ_{ext} from the BLR region. These are called synchrotron self-Compton and external Compton respectively. Others believe it is *hadronic* with e^- and p being accelerated to high energies with $p \rightarrow \gamma + \pi$. The emission is then dominated by *proton synchrotron* and external Compton. We typically observe the X-ray and γ -ray emissions to be correlated as would be predicted by leptonic models. As such these are favored.¹⁴

Due to relativistic beaming then the flux of blazars is heavily amplified.

5.4. γ -ray bursts

γ -ray *bursts* or *GRBs* are essentially extremely powerful explosions. These flashes of high-energy radiation are produced in *catastrophic events* such as neutron-star mergers. When they occur they are the brightest sources in the Universe. We distinguish between *short* and *long* GRBs.¹⁵ They are generally believed to arise from relativistic jets moving very at speeds very, very close to the speed of light. As expected we mostly observe the long duration GRBs. These are often formed when massive stars collapse and become a supernovae. Therefore supernovae are usually seen after a long GRB. The emission of GRBs is across the entire electromagnetic spectrum and is generally believed to be caused by synchrotron emissions. However, at very high-energies additional mechanisms are required such as inverse Compton scattering leading to γ -rays.

5.5. The EBL

When γ -rays travel toward us they can interact with photons on the way and pair-produce whenever enough energy is available. The wavelength which TeV γ -rays mostly interact with is given by

$$\lambda \stackrel{\text{Bethe-Heitler}}{\simeq} 1.21 \mu\text{m} \left(\frac{E}{\text{TeV}} \right)$$

¹⁴Of course exceptions exist since the Universe is annoying.

¹⁵Lasting shorter or longer than two seconds.

where the *Bethe-Heitler* cross sections peaks for $\beta_{\text{max}} \simeq 0.7$. This pair-production leads to a loss of γ -ray photons and we observe this as the spectrum being attenuated.

This implies the background radiation fields most relevant for attenuation are the optical and infrared backgrounds. These are typically referred to as the *extragalactic background light* or *EBL*. The optical background is caused by starlight while the infrared background is caused by dust absorbing starlight and re-emitting is as thermal radiation. The EBL is very hard to measure by simply counting sources and summing all contributions. We can instead use e.g. blazars and use a fit similar to

$$F(E) = \underbrace{F_{\text{intrinsic}}(E, \theta)}_{\text{given by blazar}} \overbrace{e^{-\alpha\tau(E,z)}}^{\text{given by EBL model}}$$

Which is in many ways simpler. We can therefore use observations of AGN to constrain the EBL.

6. Fermi Acceleration

We deal with plasmas in space and these typically have high conductivity. This is problematic since any electric fields are short-circuited and therefore unable to efficiently accelerate particles. However, we do observe very energetic charged particles. The mechanism we will now describe accounts for this and is called *Fermi acceleration*. The idea is that particles are able to scatter off magnetic irregularities *frozen* into clouds in the ISM.

6.1. Magnetic mirrors

The above process is sometimes referred to as *magnetic mirrors*. We assume a charged particle travels in a non-uniform magnetic field with $v \ll c$. Then the particle will not only follow helical motion around the field lines. Because of non-uniformity the particle feels a time-varying magnetic field in its instantaneous rest frame. This induces an electric field

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

The induced $\mathbf{E}$ will act to change $v_\perp$ and also the energy as

$$\begin{aligned} \Delta E_\perp &= \frac{1}{2} m \Delta v_\perp^2 \\ &= \oint q \mathbf{E} \cdot d\mathbf{l} \\ &= q \oint \nabla \times \mathbf{E} \cdot d\mathbf{S} \\ &= -q \oint \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \end{aligned}$$

We assume the magnetic field changes by a small amount during one period of one Larmor radius r_g

$$T_g = \frac{2\pi}{\omega_g}$$

with frequency

$$\omega_{g,\text{non-rel}} = \frac{qB}{m}$$

We approximate

$$-\frac{\partial B}{\partial t} \simeq \frac{\Delta B}{T_g}$$

Then

$$\begin{aligned} \Delta E_{\perp} &\simeq q \Delta B \frac{\omega_g}{2\pi} \pi r_g^2 \\ &= \frac{1}{2} m v_{\perp}^2 \frac{\Delta B}{B} \\ &= E_{\perp} \frac{\Delta B}{B} \end{aligned}$$

or

$$\frac{\Delta E_{\perp}}{E_{\perp}} = \frac{\Delta B}{B}$$

implying

$$\frac{E_{\perp}}{B} \sim \text{constant}$$

With $v_{\perp} = v \sin \alpha$ we find

$$\frac{\sin^2 \alpha}{B} \sim \text{constant}$$

Then if B changes the pitch angle α must change accordingly thereby reflecting the particle.

6.2. 2nd order Fermi acceleration

We consider particles accelerated by collisions with gas clouds acting as magnetic mirrors.

We denote the particles mass and velocity by m and $\mathbf{v}$ respectively and the clouds mass and velocity by M and $\mathbf{u}$ respectively. We will assume $M \gg m$ and $u \ll v$. The velocity v' after scattering is simply

$$\mathbf{v}' = \frac{(m - M)\mathbf{v} + 2M\mathbf{u}}{m + M}$$

$$v' \simeq -v \pm 2u$$

With $-$ corresponding to the *head-on* case. The energy after scattering is

$$E' = \frac{1}{2}m(-v \pm 2u)^2$$

implying

$$\begin{aligned} \Delta E_{1,2} &= \frac{1}{2}m(\mp uv + 4u^2) \\ &= \mp 4\frac{u}{v}E + 4\left(\frac{u}{v}\right)^2 E \\ &\simeq \pm 4\frac{u}{v}E \end{aligned}$$

We expect the following rates

$$f_{1,2} = \frac{v \pm u}{\ell}$$

with ℓ being the particle mean free path. Then

$$\begin{aligned} \frac{\Delta E}{\Delta t} &= f_1 E_1 + f_2 E_2 \\ &= \frac{8u^2}{\ell v} E \\ &\equiv \tau_F'' E \end{aligned}$$

Where we have defined the time constant τ_F'' which is second order in u . We can compute

$$\tau_F'' \sim 10^7 \text{ years}$$

So the process is slow and competes with ionisation losses meaning it is difficult to accelerate particles with low energies to sufficiently high energies. This is related to the *injection problem*.

6.3. 1st order Fermi acceleration

We can obtain higher efficiency by considering particles moving between mutually approaching clouds or if particles cross a shock front. We consider two frames: S the observer reference frame and S' the rest frame of a magnetic cloud traveling with velocity u along the x -axis. We define the *bulk* Lorentz factor as

$$\Gamma = \frac{1}{\sqrt{1 - (u/c)^2}}$$

Then

$$E' = \Gamma(E + up_x)$$

$$p'_x = \Gamma\left(p_x + \frac{u}{c^2}E\right)$$

We assume the collision is elastic so

$$E'^* = E'; \quad p'_x{}^* = -p'_x$$

with $*$ indicating after scattering. Then

$$\begin{aligned} E^* &= \Gamma(E'^* - up'_x{}^*) \\ &= \Gamma(E' + up'_x) \\ &= \Gamma^2 \left[E + up_x + up_x + \frac{u^2}{c^2}E \right] \\ &= \Gamma^2 E \left[1 + \frac{2up_x}{E} + \frac{u^2}{c^2} \right] \end{aligned}$$

We use

$$\frac{v_x}{c^2} = \frac{p_x}{E}$$

with $v_x = v \cos \theta$ and since $u \ll c$ we have

$$\Gamma^2 \simeq 1 + \left(\frac{u}{c}\right)^2$$

implying

$$\begin{aligned}
\Delta E &= E^* - E \\
&\simeq 2E \left[\frac{uv \cos \theta}{c^2} + \frac{u^2}{c^2} \right] \\
&\stackrel{v \sim c}{\simeq} \frac{2Eu \cos \theta}{c}
\end{aligned}$$

Then the particle gains energy if the collision is head-on

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

We average $\cos \theta$ over this range using $d\Omega = \sin \theta d\phi d\theta$ and a probability distribution

$$\Delta E \sim \cos \theta$$

$$\langle \cos \theta \rangle = \frac{\int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \cos^2 \theta \sin \theta}{\int_0^{2\pi} d\phi \int_{-\pi/2}^{\pi/2} d\theta \cos \theta \sin \theta}$$

The ϕ integration cancels and we substitute $\mu = \cos \theta$

$$\begin{aligned}
\langle \cos \theta \rangle &= \frac{\int_0^1 \mu^2 d\mu}{\int_0^1 \mu d\mu} \\
&= \frac{2}{3}
\end{aligned}$$

Averaging over all angles we find $\langle \cos \theta \rangle = 0$ recovering second order Fermi acceleration.

However, for $\cos \theta > 0$ we find

$$\begin{aligned}
\langle \Delta E \rangle &= \frac{4u}{3c} \langle E \rangle \equiv \eta \langle E \rangle \\
\langle E^* \rangle &= \left(1 + \frac{4u}{3c} \right) \langle E \rangle \equiv \mathcal{B} \langle E \rangle
\end{aligned}$$

We could obtain the same result by only considering head-on collisions in the previous section

$$\frac{\Delta E}{\Delta t} = f_1 E_1$$

$$\begin{aligned}
&= 4 \frac{u}{v} E \frac{u+v}{\ell} \\
&\simeq \frac{4u}{\ell} E \\
&\equiv \frac{E}{\tau'_F}
\end{aligned}$$

With

$$\tau'_F = \frac{\ell}{4u} = 2\tau''_F \frac{u}{c} \ll \tau''_F$$

We could imagine scenarios where particles only experience head-on collisions such as due to diffusive shocks. We imagine a shock wave propagating through a medium. This shock wave is characterised by a discontinuous change in characteristics of the medium such as density, velocity and temperature. We say the shock is supersonic when the shock velocity v_s is larger than the speed of sound in the medium. The shock velocity is by definition much larger than the thermal velocity of particles $v_s \gg v_1$. Within the downstream rest frame S_{down} scattering ensures the particle distribution is isotropic. However, the upstream gas as a uniform velocity v_2 in S_{down} . Then a particle crossing the shock wave will scatter head-on with the upstream gas since

$$v_2 = \frac{3}{4}v_s$$

The same thing happens within the upstream rest frame S_{up} since the downstream gas appears to have a uniform velocity v_1 in this frame. Then a particle crossing the shock wave will scatter head-on with the downstream gas since

$$v_1 = -\frac{3}{4}v_s$$

These numbers are for a monoatomic gas where $\gamma = \frac{5}{3}$ since

$$\frac{v_1}{v_2} = \frac{\gamma+1}{\gamma-1}$$

The shock velocity is v_s in the lab frame S_{obs} . Then in the rest frame of the shock S_{shock} we have

$$v_1 = -v_s; \quad v_2 = -\frac{v_s}{4}$$

Then the numbers above follow.

With

$$u = \frac{3}{4}v_s$$

we have

$$\eta = \frac{v_s}{c}$$

and

$$\mathcal{B} = 1 + \frac{v_s}{c}$$

6.4. Emitted spectrum

We have shown how Fermi acceleration could accelerate particles to high energies. We still need to show that these particles produce the power-law particle distribution we observe. We will assume a simple model.

We consider a particle gaining an amount of energy $\mathcal{B}$. The particle could escape the acceleration region. We denote the probability this happens by P_{esc} so the probability that the particle keeps accelerating is

$$P = 1 - P_{\text{esc}}$$

Then after k cycles we have

$$N = N_0 P^k$$

particles remaining with energy larger than $E = E_0 \mathcal{B}^k$. We can find

$$N(\geq E) = N_0 \left(\frac{E}{E_0} \right)^{\ln P / \ln \mathcal{B}}$$

This quantity is related to the spectrum $N(E)$ by

$$N(\geq E) = \int_E^\infty N(E') \, dE'$$

implying

$$N(E) \sim \left(\frac{E}{E_0} \right)^{\ln P / \ln \mathcal{B} - 1}$$

We know $\mathcal{B}$ so we need to compute P_{esc} . We consider an isotropic distribution with a particle flux

$$\Phi(E) = \frac{dN}{dE}$$

hitting the *planar* shock front. The flux emitted from one hemi-sphere of a source is

$$\begin{aligned} F(E) &= \int \frac{d^2 N}{dE d\Omega} \cos \theta \, d\Omega \\ &= \Phi(E) \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos \theta \sin \theta \, d\theta \\ &= \pi \Phi(E) \end{aligned}$$

Then

$$\begin{aligned} n &= \frac{4\pi}{c} \Phi(\geq E) \\ &= \frac{4}{c} F(\geq E) \end{aligned}$$

implying

$$F(\geq E) = \frac{cn}{4}$$

This is the average flux of particles through the shock front. Within S_{shock} particles can escape the acceleration region with velocity

$$v_2 = \frac{v_s}{4}$$

Then the escape flux is

$$F_{\text{esc}} = nv_2 = \frac{nv_s}{4}$$

implying

$$P_{\text{esc}} = \frac{F_{\text{esc}}}{F} = \frac{v_s}{c}$$

Then

$$P = 1 - \frac{v_s}{c}$$

and

$$\ln P \simeq -\frac{v_s}{c}; \quad \ln \mathcal{B} \simeq \frac{v_s}{c}$$

implying

$$N(E) \sim \left(\frac{E}{E_0} \right)^{-2}$$

which is reasonable!

6.5. Hillas' criterion

The rate of energy gain is roughly

$$\frac{dE}{dt} \simeq \eta E \frac{1}{T_{\text{cycle}}}$$

with T_{cycle} being the time between scatterings. We approximate

$$\lambda_{\text{cycle}} \sim r_g = \frac{p_{\perp}}{ZeB} = \frac{E}{ZeBc}$$

Then in S_{shock}

$$\begin{aligned} T_{\text{cycle}} &= \frac{r_g}{u} \\ &= \frac{E}{ZeBuc} \end{aligned}$$

and

$$\begin{aligned} \frac{dE}{dt} &\simeq \frac{\Delta E}{T_{\text{cycle}}} \\ &= \eta ZeBuc \\ &\simeq ZeBu^2 \end{aligned}$$

The maximal is

$$\begin{aligned} E_{\max} &= \frac{dE}{dt} T_{\text{shock}} \\ &= ZeBuL \end{aligned}$$

where $T_{\text{shock}} = L/u$ with L being the size of the accelerating region. We write this as

$$\frac{E_{\max}}{Z\beta_u} = eBcL$$

with $\beta_u = u/c$. This is often written in the form

$$\frac{E_{\max}}{Z\beta_u} \simeq 10^{21} \text{ eV} \left(\frac{B}{\text{G}} \right) \left(\frac{L}{\text{pc}} \right)$$

which is called *Hillas' criterion*. We see either B must be large or L must be large.

7. Galactic Sources of CRs

7.1. SNR

The best candidates for energies up to the *knee* are supernova remnants or SNRs. We consider stars with $M \gtrsim 8M_{\odot}$. These will experience *core-collapse* when fusion can no longer keep them stable.¹⁶ When collapse stops in-falling material will bounce back forming a shockwave. The now dense core quickly cools through ν -emission, with some ν being absorbed by the shell driving the shockwave.¹⁷ After the dust has settled we are left with a dense *neutron core* or neutron star surrounded by an $\sim$ spherical shell.¹⁸ Less massive stars will experience a similar core-collapse with less energy. These leave behind *white dwarfs* since they do not have sufficient mass to form a neutron core. We observe these core-collapse supernovae through ν -emissions and GWs since these events are quite extreme.¹⁹

We now show these are candidates due to 1st order Fermi acceleration. We want particles to gain energy upon crossing the shockwave, we need enough SNe with sufficient energy in their ejecta to account for the observed CR density, and the maximal energy should be high enough, per Hillas' criterion to accelerate CRs up to the knee. We also expect $\alpha \sim 2$ which we showed is possible in the last section. We estimate v_{ejecta} . Assume a star is a sphere of constant density ρ . Then

$$M = \frac{4}{3}\pi r^3 \rho \Rightarrow \frac{dM}{dr} = 4\pi r^2 \rho$$

implying

$$\begin{aligned} dU_{\text{grav}} &= \frac{GM dM}{r} \\ &= \frac{16\pi^2 Gr^4 \rho dr}{3} \end{aligned}$$

¹⁶These massive stars can not create energy through the fusion of iron thereby collapsing.

¹⁷99% of the released E_{grav} is released through these ν .

¹⁸This is caused by inverse β -decay.

¹⁹Aside from the visible explosion.

and

$$\begin{aligned}\Delta U &= \int_R^{R_{\text{NS}}} \frac{dU_{\text{grav}}}{dr} dr \\ &\simeq \frac{3GM_{\text{NS}}^2}{5R_{\text{NS}}}\end{aligned}$$

Assuming $M_{\text{NS}} = 1.4M_{\odot}$ and $R_{\text{NS}} \sim 10$ km we find

$$\Delta U \sim 10^{53} \text{ erg}$$

This is the amount of liberated energy. We assume 1% drive the shockwave while the rest escapes in the form of ν

$$K \sim 10^{51} \text{ erg}$$

implying

$$\eta = \frac{v_s}{c} \sim \frac{1}{c} \sqrt{\frac{2K}{M}} \sim 0.02$$

Which is non-relativistic but still much larger than typical v_s . The shockwave expands and sweeps up the ISM with

$$\rho_{\text{ISM}} \sim 1.6 \times 10^{-24} \frac{\text{g}}{\text{cm}^3}$$

This happens until $\rho_{\text{SN}} \sim \rho_{\text{ISM}}$ after which the shockwave slows down significantly. This is called the *Sedov phase*. This occurs when

$$\rho_{\text{SN}} = \frac{3M_{\text{SN}}}{4\pi R_{\text{SN}}^3} = \rho_{\text{ISM}}$$

With $M_{\text{SN}} \sim 10M_{\odot}$ we find

$$R_{\text{SN}} \sim 5 \text{ pc}$$

and

$$\tau_{\text{SN}} \sim \frac{R_{\text{SN}}}{v_s} \sim \mathcal{O}(10^3 \text{ yr})$$

This is a long time which can be used to accelerate particles! The typical SNe is currently in the Sedov phase. An important exception is the *Crab nebula* which is being driven continuously by a central *pulsar*. We observe 3 core-collapse SNe per century so

$$L = K f_{\text{SN}} \sim 10^{42} \frac{\text{erg}}{\text{s}}$$

which we can compare with the total CR luminosity. We have

$$\rho_{\text{CR}} \sim 1 \frac{\text{eV}}{\text{cm}^3}$$

and with $V_{\text{Galactic}} \sim 10^{66} \text{cm}^3$ and $\tau_{\text{esc}} \sim 10^7 \text{yr}$ we find

$$L_{\text{CR}} \sim \frac{\rho_{\text{CR}} V_{\text{Galactic}}}{\tau_{\text{esc}}} \sim 10^{40} \frac{\text{erg}}{\text{s}}$$

Then 1% of the energy released in SN ejecta is enough to explain the observed CR density.

The maximal energy is given by Hillas' criterion

$$E_{\text{max}} = e B c R_{\text{SN}} Z \beta_u$$

$$\stackrel{B=4\mu\text{G}}{\sim} 300 Z \text{ TeV}$$

Then SNRs can accelerate CRs up to the knee. Then SNRs are nice candidates.

When observing SNRs we see a π^0 bump at low energies. This is expected when we assume CRs are accelerated since this leads to hadronic interactions. However, the identification of PeVatron sources is weird. The detection of $\gtrsim 100 \text{ TeV}$ γ -rays would suggest the presence of freshly accelerated CR protons with PeV energies. These γ -rays would be produced in the π^0 decay produced in $2p \rightarrow \pi X$ while leptonic interactions would be suppressed at these energies. Whenever a π^0 bump is observed SNRs are generally the preferred candidate. However, PWNe can also lead to $\gtrsim 100 \text{ TeV}$ γ -rays.

7.2. White dwarfs

We would like to understand compact objects such as NS since these are observed at high energies and drive *pulsar wind nebulae*. They also emit GW by existing and through

mergers as we will see.²⁰

To understand these objects we start with the simplest being white dwarfs. These WD stop collapsing due to two quantum mechanical effects since their densities becomes immense. Those being Heisenberg's uncertainty principle and Fermi's exclusion principle. When the intermolecular spacing Δx becomes small the spread in momenta Δp becomes very small thereby providing *degeneracy pressure* to withstand further collapse. We would like to estimate this pressure. We have

$$\Delta x \Delta p \sim p \cdot d \sim \hbar$$

with each e^- occupying a cube of volume d^3 implying

$$n \sim \frac{1}{d^3}$$

and

$$p \sim \frac{\hbar}{d} \sim \hbar n^{1/3}$$

Assuming the particles are non-relativistic we have

$$E = \frac{p^2}{2m_e} \sim \frac{\hbar^2 n^{2/3}}{2m_e}$$

This is basically the *Fermi energy*. A proper treatment gives

$$E_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}$$

Consider an ionized hydrogen gas with

$$\rho \sim \frac{m_p}{d^3}$$

and

$$\varepsilon = \frac{E\rho}{m_p}$$

²⁰BH binaries can also form *micro-quasars* etc.

$$= \frac{\hbar^2}{2m_e} \left(\frac{\rho}{m_p} \right)^{5/3}$$

We assume the gas is ideal and adiabatic

$$p \sim \rho^\gamma$$

where

$$\gamma = \left(\frac{\partial \ln p}{\partial \ln \rho} \right)_S$$

This follows from

$$pV^\gamma = \text{constant}$$

implying

$$\frac{dp}{p} = -\gamma \frac{dV}{V}$$

and by definition of ρ

$$\frac{d\rho}{\rho} = -\frac{dV}{V}$$

Also,

$$p = (\gamma - 1)\varepsilon$$

which follows from $dU = -p dV$. Then

$$p \sim \varepsilon \sim \rho^{5/3}$$

implying

$$\gamma = \frac{5}{3}$$

and

$$p = \frac{2}{3}\varepsilon \sim \frac{\hbar^2}{3m_e} \left(\frac{\rho}{m_p} \right)^{5/3}$$

A proper treatment gives

$$p = \frac{(3\pi^2)^{2/3}}{5} \frac{\hbar^2}{m_e} \left(\frac{\rho}{\mu_e m_u} \right)^{5/3} \sim \rho^{5/3}$$

which is valid for all compositions. We would like to know what happens to the equation of state when e^- become relativistic, and when this occurs. We can write

$$\Delta p \stackrel{\text{relativistic}}{\sim} m_e c$$

implying

$$\rho_{\text{crit}} = \frac{m_p}{d^3} \sim m_p \left(\frac{m_e c}{\hbar} \right)^3 \sim 3 \times 10^{10} \frac{\text{kg}}{\text{m}^3}$$

A proper treatment gives²¹

$$\rho_{\text{crit}} = \frac{m_u \mu_e}{3\pi^2} \left(\frac{m_e c}{\hbar} \right)^3 \sim 10^9 \mu_e \frac{\text{kg}}{\text{m}^3}$$

We can determine the equation of state as before. However, now we use

$$E = \sqrt{p^2 c^2 - (m c^2)^2} \sim p c \sim \frac{\hbar c}{d} \sim \hbar c n^{1/3}$$

implying

$$\varepsilon \sim \frac{E}{d^3} \sim \frac{\hbar c}{d^4}$$

with

$$\rho \sim \frac{m_p}{d^3}$$

we find

$$\varepsilon \sim \hbar c \left(\frac{\rho}{m_p} \right)^{4/3}$$

implying

$$\gamma = \frac{4}{3}$$

and

$$p = \frac{\varepsilon}{3} \sim \frac{\hbar c}{3} \left(\frac{\rho}{m_p} \right)^{4/3}$$

A proper treatment gives

²¹We require $\hbar k_F = m_e c$

$$p = \frac{(3\pi^2)^{1/3}}{4} \hbar c \left(\frac{\rho}{\mu_e m_u} \right)^{4/3} \sim \rho^{4/3}$$

These two equations are the equations of state for degenerate e^- gasses, including those within WDs and NS.

We can balance the pressure of an e^- gas like those above with U_{grav} to determine the maximal allowed mass of a WD.²² We apply the *virial theorem* to get a rough estimate

$$2U = U_{\text{grav}}$$

with

$$U = \varepsilon V \sim \hbar c \underbrace{\left(\frac{\rho V}{m_p} \right)^{4/3}}_{(M/m_p)^{4/3}} \underbrace{V^{-1/3}}_{\sim R^{-1}}$$

$$U_{\text{grav}} = \frac{1}{2} \frac{GM^2}{R}$$

implying

$$M_{\text{Ch}} \sim \left(\frac{\hbar c}{G} \right)^{3/2} \frac{1}{m_p^2} \sim 2M_{\odot}$$

which we call the *Chandrasekhar mass*. We can define a *gravitational fine structure constant*

$$\alpha_G = \frac{Gm_p^2}{\hbar c} \sim 10^{-39}$$

and write

$$M_{\text{Ch}} \sim m_p \alpha_G^{-3/2}$$

implying a WD consists of $\sim 10^{60}$ protons. A proper treatment gives

$$M_{\text{Ch}} \sim 1.4M_{\odot}$$

We can now estimate the size of a WD

$$R_{\text{WD}} \sim \left(\frac{M_{\text{Ch}}}{\rho_{\text{crit}}} \right)^{1/3} = \frac{\lambda_e}{\sqrt{\alpha_G}} \sim \frac{R_{\odot}}{100}$$

²²or NS.

where λ_e is the *Compton wavelenght* of e^-

$$\lambda_e = \frac{\hbar}{m_e c}$$

7.3. NS and pulsars

We now consider NS. As core-collapse occurs we have stages of increasing ρ . The following occur in order

1. e^- gas becomes degenerate $p \sim \rho^{5/3}$.
2. e^- gas becomes relativistic $p \sim \rho^{4/3}$ at ρ_{crit} .
3. At $\rho \sim 10^{10} \frac{\text{kg}}{\text{m}^3}$ we have *neutronisation*. The e^- have enough energy to perform inverse β -decay: $p + e^- \rightarrow n + \nu_e$

$$E_{\text{tot}} = \gamma m_e c^2 \geq (m_n - m_p) c^2$$

The neutrons cannot perform β -decay as all e^- states are occupied. This occurs for

$$p_F \geq \sqrt{E_{\text{tot}}^2/c^2 - m_e^2 c^2}$$

4. At $\rho \sim 10^{14} \frac{\text{kg}}{\text{m}^3}$ we have the *neutron drip process*. The heavy nuclei become enriched with neutrons from inverse β -decay making them unstable.
5. At $\rho \sim 10^{17} \frac{\text{kg}}{\text{m}^3}$ we have a degenerate n gas. This occurs after all $p \rightarrow n$. All equations for NS are the same as those for WD “derived” above.

We note that the magnetic field of a star is conserved during core-collapse. The surface magnetic field B scales as

$$B \sim r^{-2}$$

implying NS can have very large magnetic fields. We also observe NS that rotate rapidly. These are called *pulsars*. We observe them as very stable, short periods of pulses and polarized radio emission. The idea is that radio emission is along a magnetic axis which is misaligned with the rotational axis. This would lead to pulses as we observe.

We can estimate the surface magnetic field using the pulse period P and the *spin-down rate* $\dot{P}$. The rotation slows down with *braking index* n

$$\dot{\Omega} = -\kappa\Omega^n$$

where Ω is the angular frequency and n provides information about which processes act to slow the rotation. The important braking mechanism for our purposes is *magnetic braking*. The idea is that a magnetic dipole forms at an angle with respect to the rotational axis. This leads to a varying dipole moment and thereby radiation. We can use the *Larmor formula* again

$$-\frac{dE}{dt} = \frac{\mu_0 |\ddot{p}_m|^2}{6\pi c^3}$$

a rotating dipole has

$$p_m = p_{m0} \sin \Omega t$$

Then

$$\left\langle -\frac{dE}{dt} \right\rangle = \frac{\mu_0 \Omega^4 p_{m0}^2}{12\pi c^3}$$

The rotational energy is given by

$$E = \frac{1}{2} I \Omega^2$$

implying

$$-\frac{dE}{dt} = -I \dot{\Omega} \Omega$$

With the previous

$$\dot{\Omega} \sim \Omega^3$$

so $n = 3$. We can find

$$n = \frac{\Omega \ddot{\Omega}}{\dot{\Omega}^2} = \frac{\nu \ddot{\nu}}{\dot{\nu}^2} = 2 - \frac{\ddot{P}P}{\dot{P}^2}$$

Assuming n is constant during the pulsars lifetime τ we can find

$$\tau = \frac{P}{(n-1)\dot{P}} \stackrel{n=3}{=} \frac{P}{2\dot{P}}$$

With

$$B_s \sim \frac{\mu_0 p_{m0}}{4\pi R^3}$$

we can also find

$$B_s = \left(\frac{3\mu_0 c^3 M}{80\pi^3 R^4} \right)^{1/2} \sqrt{P\dot{P}}$$

where we used

$$I = \frac{2}{5}MR^2$$

We typically plot pulsars in a $P-\dot{P}$ *diagram* which can be treated as the *Hertzsprung-Russell diagram* for pulsars.

The rotating magnetic field of any pulsar induces an electric field much stronger the gravitational field. This leads to charges being ripped of the surface. These charges will rearragne such that no Lorentz force acts on them

$$0 = \mathbf{E} + (\mathbf{\Omega} \times \mathbf{r}) \times \mathbf{B}$$

The distribution of charges is given by

$$\rho = \frac{\nabla \cdot \mathbf{E}}{4\pi} \sim -\frac{\mathbf{\Omega} \cdot \mathbf{B}}{2\pi c} = \frac{B_s}{2\pi c} \left(\frac{R}{r} \right)^3 (3 \cos^2 \theta - 1)$$

The charges cannot co-rotate with field lines beyond the *light cylinder* where $R_{\text{LC}}\Omega = c$.

Any field lines that would close beyond R_{LC} become open and the charges can escape.

The last closed field line has angle

$$\sin \theta_{\text{PC}} = \sqrt{\frac{\Omega R}{c}}$$

This defines the *polar cap region* on the surface from which particles can escape to infinity.

There is a large potential difference between the pole and the edge of the polar cap region given by

$$-\Delta\Phi \sim \frac{1}{2}B_0 R \theta_{\text{PC}} \sim 10^{13} \text{V}$$

When observing pulsars we see radiation across the entire spectrum. The produced emission is believed to originate due to charged particles being accelerated along field lines i.e. through synchrotron radiation. This is also believed to cause cascades like in our atmosphere. There is likely also some IC emission.

7.4. PWNe

When a pulsar is embedded in a nebula from some SN explosion then the spin-down luminosity from said pulsar can drive emission from the nebula. The e^- from the pulsar can be accelerated to PeV energies by the SNR shockwave. These are referred to as *pulsar wind nebulae*.²³ Also, see TeV *halos*.

²³An example of such an object is the *Crab nebula*.

8. Gravitational Waves

The existence of GWs was proposed by Poincaré in 1905. Using relativity Einstein and Rosen concluded in 1936 that GWs did not exist. Robertson²⁴ refereed their paper and found a mistake changing the conclusion. The first direct detection of GWs was published in 2015. However, searches has been done since the 1960s and the first indirect observion was done through the *Hulse-Taylor system*. This system is a NS binary with one NS being a pulsar. The measured change in orbital period $\dot{P}$ was consistent with the emission of GWs.

8.1. Basics of GWs

According to relativity space is a four-dimensional manifold with gravity being caused by the curvature of the manifold. The object of interest in relativity is the metric $g_{\mu\nu}$ defined by

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu$$

Minkowski spacetime has $g_{\mu\nu} = \eta_{\mu\nu}$ with $\eta_{\mu\nu}$ being the Minkowski metric. When spacetime is not Minkowski the metric can be much more complicated. An example of such a metric is the *Friedmann-Robertson-Walker metric* describing a homogeneous and isotropic Universe

$$ds^2 = dt^2 - \frac{a^2(t)}{c^2} d\Sigma^2$$

with

$$d\Sigma^2 = \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2$$

The metric is related to the energy-momentum tensor $T_{\mu\nu}$ by the *Einstein field equations*

$$\mathcal{R}_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\mathcal{R} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

We consider a non-static source. Away from the source spacetime should be $\sim$ Minkowski with small distortions. Then

²⁴Of Friedmann-Robertson-Walker fame.

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x)$$

with $h_{\mu\nu}$ small. We plug this metric into the EFE with $T_{\mu\nu} = 0$. Then ignoring higher order terms we have

$$\square h_{\mu\nu} = 0$$

which is a wave-equation! We make the ansatz

$$h_{\mu\nu} = h_{\mu\nu}^0 e^{ik_\mu x^\mu} = h_{\mu\nu}^0 e^{i\mathbf{k}\cdot\mathbf{x}} e^{-i\omega t}$$

where $h_{\mu\nu}^0$ is a symmetric 4×4 matrix and $\omega = kc$. Let the propagation be along $\hat{z}$. Then in the *transverse-traceless gauge* we have

$$h_{\mu\nu}^0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

with h_+ and $h_\times$ being constant amplitudes. With $T_{\mu\nu} \neq 0$ we find the inhomogeneous wave-equation

$$\square h_{\mu\nu} = \frac{16\pi G}{c^4} T_{\mu\nu}$$

which can be solved using Green's functions

$$h_{\mu\nu}(x) = \frac{4G}{c^4} \int \frac{d^3x'}{|\mathbf{x} - \mathbf{x}'|} T_{\mu\nu}(t', x')$$

where we evaluate $T_{\mu\nu}$ at the *retarded time*

$$t' = t - \frac{|\mathbf{x} - \mathbf{x}'|}{c}$$

We will consider sources of size R that vary harmonically over time with frequency ν_s .

Also, we assume $T_{\mu\nu}$ is dominated by the rest mass of the rotating objects whose density is ρ_m . We also need some approximations

$$\text{long wavelength approx. } \lambda \gg R$$

$$\text{far field approx. } r \gg R$$

With these we have

$$h_{\mu\nu}(t, \mathbf{x}) \simeq \frac{4G}{rc^4} \int d^3x' T_{\mu\nu}\left(t - \frac{r}{c}, \mathbf{x}'\right)$$

Then

$$h_{ij} \simeq \frac{4G}{rc^4} \frac{d^2 Q_{ij}}{dt^2}$$

with Q_{ij} being the *mass quadropole moment*

$$Q_{ij} = \int d^3x \left(x_i x_j - \frac{1}{3} r^2 \delta_{ij} \right) \rho_m(\mathbf{x})$$

8.2. Energy of GWs

Any GW will distort the distance L between masses by ΔL . We quantify the distortion using the *strain* h defined by

$$h = \frac{\Delta L}{L}$$

which will oscillate with the frequency of the GW.

The GW radiation is quadropolar to lowest order. There is no monopole- or dipole radiation due to energy conservation and momentum conservation respectively.²⁵ Also, GW radiation is not spherically symmetric just like EM radiation. The GW flux in a volume $V \sim L^3$ with $L \gg \lambda$ is given by

$$F = \frac{1}{32\pi} |\dot{h}|^2 \frac{c^3}{G}$$

and the total luminosity is given by

$$L = \frac{G}{5c^5} \sum_{i,j=1}^3 \left(\frac{d^3 Q_{ij}}{dt^3} \right)^2$$

Note, as opposed to EM radiation GWs with $\nu \lesssim \text{kHz}$ come from sources with sizes $R \lesssim \lambda$ implying the signal is *coherent*. Also, the scaling is more favorable as

$$\text{EM flux} \sim \frac{1}{r^2}$$

²⁵Since dipole radiation is associated with movement of the CM.

while

$$\text{strain} \sim \frac{1}{r}$$

Then GWs are detectable at $\nu \lesssim \text{kHz}$, however, *graviton* energies $\hbar\omega$ are small and their detection therefore seems unfeasible. GWs also propagate almost freely allowing us to in principle detect GWs from the very early Universe. Thereby we can probe beyond the CMB.

8.3. Binary mergers

We consider two masses m_1 and m_2 in a binary orbit with orbital frequency ω_s .²⁶ The angular velocity is related to the orbit size R by

$$\omega_s^2 = \frac{GM}{R^3}$$

with

$$M = m_1 + m_2$$

The geometry we consider is shown in Figure 1.

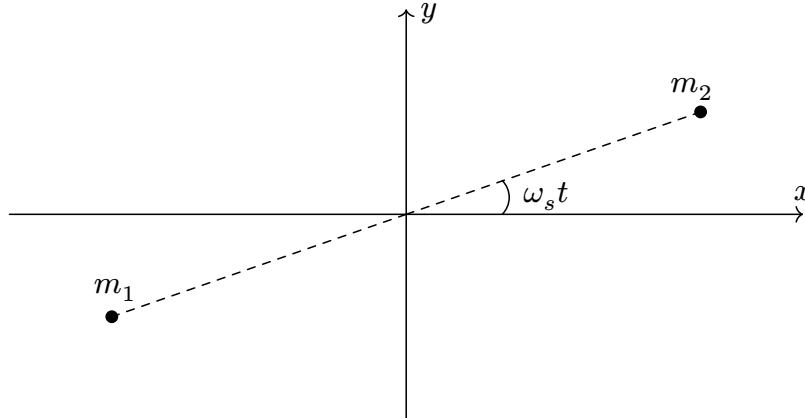


Figure 1: Geometry for the binary system.

Also, denote the distance from $(0, 0)$ to m_i by r_i . We have

$$\rho_m(\mathbf{x}) = \sum_{\alpha=1}^2 m_{\alpha} \delta(\mathbf{x} - \mathbf{x}^{\alpha})$$

implying

²⁶We will take their velocities to be non-relativistic.

$$\begin{aligned}
Q_{ij} &= \int d^3x \left(x_i x_j - \frac{1}{3} r^2 \delta_{ij} \right) \rho_m(\mathbf{x}) \\
&= \sum_{\alpha=1}^2 m_{\alpha} \left(x_i^{\alpha} x_j^{\alpha} - \frac{1}{3} r_{\alpha}^2 \delta_{ij} \right)
\end{aligned}$$

We have $z = 0$ for m_{α} so

$$Q_{33} = \sum_{\alpha=1}^2 m_{\alpha} \left(-\frac{1}{3} r_{\alpha}^2 \right)$$

and

$$Q_{3i} = Q_{i3} = 0$$

Also,

$$\begin{aligned}
Q_{11} &= \sum_{\alpha} m_{\alpha} \left((x_1^{\alpha})^2 - \frac{1}{3} r_{\alpha}^2 \right) \\
&= \sum_{\alpha} m_{\alpha} \left(\frac{2}{3} (x_1^{\alpha})^2 - \frac{1}{3} (y_2^{\alpha})^2 \right)
\end{aligned}$$

and

$$Q_{22} = \sum_{\alpha} m_{\alpha} \left(\frac{2}{3} (y_1^{\alpha})^2 - \frac{1}{3} (x_2^{\alpha})^2 \right)$$

$$Q_{12} = Q_{21} = \sum_{\alpha} m_{\alpha} x^{\alpha} y^{\alpha}$$

With $R = r_1 + r_2$ we can write

$$Q_{ij} = \frac{1}{2} \mu R^2 J_{ij}$$

with

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

and

$$J_{ij} = \begin{pmatrix} \cos 2\omega_s t + \frac{1}{3} & \sin 2\omega_s t & 0 \\ \sin 2\omega_s t & \frac{1}{3} - \cos 2\omega_s t & 0 \\ 0 & 0 & -\frac{2}{3} \end{pmatrix}$$

We can approximate

$$J_{ij} \sim \cos 2\omega_s t; \quad Q_{ij} \sim Q; \quad h_0 \sim h_+ \sim h_\times$$

so

$$\frac{d^2 Q}{dt^2} = \frac{1}{2} \mu R^2 (4\omega_s^2) \cos 2\omega_s t$$

implying

$$\begin{aligned} h(t) &\sim \frac{4G}{rc^4} 2\mu R^2 \omega_s^2 \cos \omega_{\text{GW}} t \\ &= h_0 \cos \omega_{\text{GW}} t \end{aligned}$$

with $\omega_{\text{GW}} = 2\omega_s$. We can rewrite h_0 as

$$\begin{aligned} h_0 &\sim \frac{4G}{rc^4} 2\mu R^2 \omega_s^2 \\ &= \frac{4G}{rc^4} 2\mu R^2 \frac{GM}{R^3} \\ &= 2 \left(\frac{2GM}{c^2 r} \right) \left(\frac{2G\mu}{c^2 R} \right) \\ &= \frac{2}{rR} \left(\frac{2Gm_1}{c^2} \right) \left(\frac{2Gm_2}{c^2} \right) \\ &= \frac{2R_1 R_2}{rR} \end{aligned}$$

where R_i are the masses Schwarzschild radii! As an example consider a binary NS system

with $m_1 \sim m_2 \sim 1.4M_\odot$ so $R_i \sim 4\text{km}$. With $r = 40\text{Mpc}$ and $R = 100\text{km}$ we find

$$h_0 \sim 10^{-22} \sim \frac{\Delta L}{L}$$

which is tiny.²⁷

We can find the luminosity by computing $\ddot{Q}_{ij}$ and we find

$$L = \frac{32G}{5c^5} (\mu R^2 \omega_s^3)^2$$

²⁷This is why constructing GW detectors is hard. One needs to take extreme care to remove noise.

and the flux is

$$F \sim 10^{-5} \frac{\text{W}}{\text{m}^2}$$

for $\omega_s \sim 400$ Hz. This is a huge amount of flux!

8.4. The inspiral stage

The energy of the binary system is given by

$$\begin{aligned} E &= \frac{1}{2} \mu \omega_s^2 R^2 - \frac{G m_1 m_2}{R} \\ &= -\frac{1}{2} m_1 m_2 \frac{G}{R} \\ &= -\frac{G \mu M}{2R} \end{aligned}$$

During the inspiral stage GW are emitted leading to orbital decay. The energy loss is

$$\begin{aligned} -\frac{dE}{dt} &= -\frac{1}{2} \frac{G \mu M}{R} \frac{\dot{R}}{R} \\ &= \frac{1}{3} G \mu M \frac{\dot{\omega}_s}{\omega_s} \\ &\stackrel{!}{=} L = \frac{32G}{5c^5} \mu^2 R^4 \omega_s^6 \end{aligned}$$

We eventually find

$$\dot{\omega}_s^3 = \left(\frac{96}{5}\right)^3 \frac{\omega_s^{11}}{c^{15}} (G m_c)^5$$

where we define the *chirp mass* m_c by

$$m_c = (\mu^3 M^2)^{1/5} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$

which is the characteristic scale of the inspiral stage. We can use $\omega_s = \pi \nu_{\text{GW}}$ to obtain

$$m_c = \frac{c^3}{G} \left[\frac{5}{96} \pi^{-8/3} \nu_{\text{GW}}^{-11/3} \dot{\nu}_{\text{GW}} \right]^{3/5}$$

Then we can compute m_c by knowing ν_{GW} and $\dot{\nu}_{\text{GW}}$! This is done in practice by using

$$\nu_{\text{GW}} = \frac{1}{\Delta t}; \quad \dot{\nu}_{\text{GW}} = \frac{\Delta \nu_{\text{GW}}}{\Delta t}$$

8.5. The merger stage

When *coalescence* or merging begins our Newtonian approximation breaks down horribly. We expect this happens when the orbital radius corresponds to the Schwarzschild radius of the combined masses

$$R_s = \frac{2GM}{c^2}$$

implying

$$\omega_{\text{max}} = \frac{1}{\sqrt{8}} \frac{c^3}{GM} \sim 2 \times 10^5 \text{ Hz} \left(\frac{M_{\odot}}{M} \right)$$

This frequency should correspond to the maximally observed GW frequency

$$\omega_{\text{max}} \simeq \pi \nu_{\text{GW}}^{\text{max}}$$

We can then measure $\nu_{\text{GW}}^{\text{max}}$ to compute the total mass M . With the chirp mass m_c we can now determine m_1 and m_2 . We define the *mass ratio* q by

$$m_1 = qM; \quad m_2 = (1 - q)M$$

implying

$$M = \frac{m_c}{(q(1 - q))^{3/5}}$$

Then q can be inferred!

8.6. Luminosity distance

The *luminosity distance* d_L is defined by

$$F = \frac{L}{4\pi d_L^2}$$

Using the above equations for F and L we find

$$d_L = \frac{8G}{\sqrt{5}c^4} \frac{1}{h_0} (\mu R^2 \omega_s^4)$$

We can use this to probe the *Hubble constant* H_0 since

$$d_L(z_0) = (1 + z_0)c \int_0^{z_0} \frac{dz}{H(z)}$$

Then knowing z_0 allows us to probe these cosmological parameters using GWs. This is likely not viable for BH mergers since there is no EM counterpart to measure z_0 . However, we can apply this to NS mergers which have been observed!

Also, GWs are redshifted over cosmological distances as

$$\nu_{\text{GW}}^{\text{obs}} = \frac{\nu_{\text{GW}}^{\text{emit}}}{1 + z}$$

and

$$\begin{aligned} \dot{\nu}_{\text{GW}}^{\text{obs}} &= \frac{d\nu^{\text{obs}}}{dt^{\text{obs}}} = \underbrace{\frac{d\nu^{\text{obs}}}{d\nu^{\text{emit}}}}_{(1+z)^{-1}} \underbrace{\frac{d\nu^{\text{emit}}}{dt^{\text{emit}}}}_{(1+z)^{-1}} \frac{dt^{\text{emit}}}{dt^{\text{obs}}} \\ &= \frac{\dot{\nu}_{\text{GW}}^{\text{emit}}}{(1 + z)^2} \end{aligned}$$

implying the chirp mass is also redshifted as

$$m_c^{\text{obs}} = (1 + z)m_c^{\text{emit}}$$

and similarly for m_1^{obs} and m_2^{obs} .

8.7. Spinning black holes

The spin of BHs leads to velocity-dependent interaction during inspiral which are annoying to deal with analytically. Therefore such effects are typically handled numerically and by fitting data.

We consider a massive object of mass m and spin $\mathbf{S}$. Then we define a dimensionless *spin parameter*

$$\chi = \frac{c|\mathbf{S}|}{Gm^2}$$

Spinning or Kerr BHs have smaller horizons as compared to Schwarzschild BHs which have $\chi = 0$. When $\chi = 1$ one can show

$$R_{\text{horizon}} = \frac{Gm}{c^2} = \frac{R_{\text{Schwarzschild}}}{2}$$

The χ of the final BH can be determined analytically by modelling the *ringdown stage* as a damped oscillator. As an example in GW150914 the more massive BH had $\chi < 0.7$ and it was found that the final BH had

$$\chi \simeq 0.67$$

in agreement with the spin of the *primary* BH before the merger.

8.8. Pulsar timing arrays

The interferometers we use to measure GWs are not sensitive to GWs with frequencies below $\sim 10\text{Hz}$. These frequencies would occur for mergers of SMBHs with $M \gtrsim M_{\odot}$. However, such mergers could cause a *stochastic GW background* which we can measure using *pulsar timing arrays*.

Using PTAs is based on the observation that millisecond pulsars have extremely stable rotation periods

$$\nu(t) = \nu_0 + \dot{\nu}_0 t$$

with

$$\frac{\dot{\nu}_0}{\nu_0} \sim 10^{-20}\text{Hz}$$

As EM radiation propagates through spacetime where the metric is perturbed due to GWs it accumulates a time delay ΔT . We consider a pulsar a distance L away from us. Then

$$\Delta T(t) = \frac{1}{2} \hat{p}^i \hat{p}^j \int_0^L ds h_{ij}(t(s), \mathbf{x}(s))$$

where $\hat{p}$ is “along” L and

$$t(s) = t - (L - s); \quad \mathbf{x} = (L - s)\hat{p}$$

with $c = 1$. We consider a binary merger a distance d away. We define $\hat{\Omega}$ to be along d toward us and denote the wavelength of the produced GWs by λ . We can rewrite the above in terms of redshift

$$\begin{aligned}
z(t) &= \frac{d\Delta T(t)}{dt} \\
&= \frac{1}{2} \hat{p}^i \hat{p}^j \int_0^L ds \frac{\partial h_{ij}}{\partial t}(t(s), \mathbf{x}(s))
\end{aligned}$$

We consider pairs of pulsars call each member of a pair a and b . The time averaged product over an observation time T is then

$$\begin{aligned}
\rho_{ab} &= \overline{z_a(t) z_b(t)} \\
&= \frac{1}{T} \int_{-T/2}^{T/2} dt z_a(t) z_b(t)
\end{aligned}$$

with

$$z_a(t) = \frac{1}{2} \hat{p}_a^i \hat{p}_a^j \int_0^{L_a} ds \frac{\partial h_{ij}}{\partial t}(t - (L_a - s), (L_a - s) \hat{p}_a)$$

and similarly for $z_b(t)$. We average over all pairs of pulsars a and b separated by an angle γ on the sky

$$\begin{aligned}
\Gamma(\gamma) &= \langle \rho_{ab} \rangle_{a,b \in \gamma} \\
&= [h^2 \mu_u(\gamma) (1 + \delta_{ab})]_{a,b \in \gamma}
\end{aligned}$$

where μ_u is the *Hellings and Downs function*. Then comparing $\Gamma(\gamma)$ with observations could show the existence of a GW background. The explanations for the GW background is usually mergers of SMBH or sources from the very early Universe such as inflation!

9. Evidence for Dark Matter

9.1. The Coma cluster

Fritz Zwicky showed in 1933 by applying the virial theorem that visible matter was not enough to account for the observed dynamics of the Coma cluster.

We retrace Zwicky's calculation. By the virial theorem

$$2\langle E_{\text{kin}} \rangle = -\langle E_{\text{pot}} \rangle$$

We assume the Coma cluster is spherical with

$$\langle E_{\text{pot}} \rangle = -\alpha \frac{GM^2}{r}$$

where $\alpha \sim$ density profile and M is the total mass. Computing $\langle E_{\text{kin}} \rangle$ requires knowing the average velocity relative to the COM in the Coma cluster. We use the dispersion

$$\sigma^2 = \langle v^2 \rangle - \langle v \rangle^2$$

However, we can only measure $\sigma_{\parallel}$ so we need to assume the Coma cluster is isotropic implying

$$\sigma = \sqrt{3}\sigma_{\parallel}$$

Also, assuming the Coma cluster is uniform we have

$$\alpha = \frac{3}{5}$$

Then by the virial theorem

$$\sigma_{\parallel}^2 = \frac{GM}{5R}$$

with R being the radius of the Coma cluster. Zwicky measured seven galaxies in the Coma cluster and found

$$\sigma_{\parallel} \sim 1000 \frac{\text{km}}{\text{s}}$$

Likewise, he measured

$$R \sim 1 \text{ Mpc}$$

Then

$$M_{\text{virial}} \sim 10^{14} M_{\odot}$$

Also, taking the luminosity of the Coma cluster to be

$$L_{\text{Coma}} \sim [10^{12}, 10^{13}] L_{\odot}$$

Assuming the Coma cluster has $N_{\text{Coma}} = 800$ galaxies with $M_{\text{Galaxy}} \sim 10^9 M_{\odot}$ we find

$$M_{\text{count}} \sim 10^{12} M_{\odot}$$

This is much less than M_{virial} and we can define the *mass-to-light-ratio*

$$\frac{M_{\text{virial}}}{L_{\text{Coma}}} \sim 5 \cdot 10^2$$

We can make much more conservative estimates and still miss $\sim 80\%$ of the mass.

This was the $\sim$ first evidence for dark matter.

9.2. Rotation curves

Vera Rubin studied the velocities of stars in galaxies in the 70s and 80s deriving their rotation curves $v(r)$

$$v(r) = \sqrt{\frac{GM(r)}{r}}$$

We observe almost all the luminous matter in a galaxy to be toward the center bulge. Therefore at large r we would expect $M(r) \sim \text{constant}$ and $v \sim r^{-1/2}$. However, Rubin found $v(r) \sim \text{constant}$ for large r implying $M(r) \sim r$ and $\rho \sim r^{-2}$. Then galaxies appear to be surrounded by a halo of non-luminous dark matter. This is shown to very large r using the 21cm line.

One of the most commonly used density profiles for the dark matter halo is the Navarro-Frenk-White profile

$$\rho_{\text{NFW}}(r) = \rho_s \frac{r_s}{r} \left(1 + \frac{r_s}{r}\right)^{-2}$$

Just to give an example.

9.3. N-body simulations

Modern *proof* of dark matter. Using simulations people have shown we need dark matter to achieve proper structure formation due to the clumping nature of dark matter.

9.4. Gravitational lensing

We know from relativity that matter can deform spacetime and therefore bend light. Then a large enough mass will act as a lens. This could be a quasar, etc. or more interestingly dark matter. We can use this to learn things about the source or attempt to reconstruct the lens.

An example where this has been used is the Bullet cluster where dark matter is needed to explain the amount of gravitational lensing. The Bullet cluster consists of two clusters colliding. When this happens the dark matter halos of the cluster pass through each other uninhibited while clouds of gas are slowed due to friction. This acts to separate the luminous matter and dark matter from which one can show the lensing happens mostly where there is no visible matter.

9.5. MOND

Modified Newtonian dynamics or MOND attempts to explain what dark matter does without dark matter. MOND assumed Newtonian dynamics fail at small accelerations. This is modelled by

$$\mu\left(\frac{a}{a_0}\right)a = -|\nabla\phi_N|$$

where μ is a function going to 1 as $a \gg a_0$ with $a_0 \sim 10^{-10} \text{ m s}^{-2}$ and

$$\phi_N = -\frac{GM}{r}$$

With rotation curves we have $a \ll a_0$ at large r so

$$\mu\left(\frac{a}{a_0}\right) \sim \frac{a}{a_0}$$

and

$$\frac{a^2}{a_0} = \frac{GM}{r^2} \Rightarrow a^2 = a_0 \frac{GM}{r^2}$$

Using $a = v^2/r$ we find

$$v = (a_0 GM)^{1/4} \sim \text{constant}$$

so MOND leads to $v \sim \text{constant}$ at large r , however, there is little else it explains and it has no real motivation. Briefly, it is stupid.

10. Λ CDM

10.1. Hubble's law

Hubble's law is the observation that galaxies move away from us with a velocity given by

$$v = H_0 d$$

where

$$H_0 \sim 68 \text{ km s}^{-1} \text{Mpc}^{-1}$$

is the Hubble constant.

Within relativity this is a consequence of the Universe and spacetime expanding isotropically and homogeneously. Under these assumptions any observer would see Hubble like behaviour. We write

$$d = a(t)\chi$$

where χ is the comoving distance between objects measured at $t = t_0$ and $a(t)$ is the scale factor normalised as $a(t_0) = 1$. Then

$$v \equiv \dot{d} = \dot{a}\chi = \frac{\dot{a}}{a}d = H(t)d$$

with

$$H(t) \equiv \frac{\dot{a}}{a}$$

being the Hubble parameter where $H_0 = H(t_0)$. We see $H(t)$ is a measure of the expansion rate of the Universe.

10.2. The FRW cosmology

We would like equations describing the expansion of the Universe more qualitatively. We make three assumptions:

1. We can use relativity.
2. The cosmological principle holds.²⁸
3. There exists a cosmic time.

Under these we can derive the Friedmann-Robertson-Walker metric

$$ds^2 = dt^2 - a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]$$

where k is related to spatial curvature.

With this metric and assuming the Universe is a perfect fluid one finds the two Friedmann equations

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3} \rho + \frac{\Lambda}{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}$$

which can be combined to give

$$\frac{d}{dt}(\rho a^3) = -p \frac{d}{dt}(a^3)$$

We need an equation of state $P(\rho)$ to solve the Friedmann equations. We will typically assume

$$P = w\rho$$

implying

$$\rho \sim a^{-3(w+1)}$$

We see for non-relativistic matter where $w = 0$ we find

$$\rho_m \sim \frac{1}{a^3}$$

²⁸This is just isotropy and homogeneity at large scales.

while for relativistic matter where $w = \frac{1}{3}$ we find

$$\rho_r \sim \frac{1}{a^4}$$

and for $\rho_\Lambda \sim \text{constant}$ we need $w = -1$. Also, we can show

$$a(t) \sim t^{2/3(w+1)}$$

Consider a de-Sitter Universe i.e. one dominated by $\Lambda > 0$. Then

$$a(t) \sim e^{Ht}$$

with

$$H = \sqrt{\frac{\Lambda}{3}}$$

Because different types of matter evolve differently the Universe has experienced various periods of domination and equality. This is obvious by considering the above scalings.

10.3. Typical form

Assume $\Lambda = 0$ then

$$\frac{k}{a^2} = \frac{8\pi G}{3}\rho - H^2$$

we define

$$\rho_{\text{crit}} = \frac{3H^2}{8\pi G}$$

which corresponds to ρ_{Universe} if $k \stackrel{!}{=} 0$. We define density parameters

$$\Omega_i = \frac{\rho_i}{\rho_{\text{crit}}}$$

and

$$\Omega_k = -\frac{k}{H^2 a^2}$$

Then the 1st Friedmann equation becomes

$$\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1$$

We could also write

$$\frac{H^2}{H_0^2} = \Omega_{0,\Lambda} + \frac{\Omega_{0,k}}{a^2} + \frac{\Omega_{0,m}}{a^3} + \frac{\Omega_{0,r}}{a^4}$$

Note, we sometimes use $1 + z = a^{-1}$.

We would now like to measure the density parameters!

10.4. The CMB

We know the CMB is a *perfect* blackbody with

$$T_{\text{CMB},0} \simeq 2.73$$

Within the Big Bang model we believe the CMB to originate when the Universe cooled sufficiently so p and e^- could form hydrogen

$$p + e^- \rightarrow H + \gamma \sim \text{recombination}$$

When recombination occurred the temperature was $T_{\text{CMB,rec}} = 3700$ K implying

$$z_{\text{rec}} \sim 1300$$

During this epoch γ could not propagate freely due to Thomson scattering with free e^- .

We have

$$\Gamma_T \sim n_e \sigma_T c \sim X(z) n_B \sigma_T c$$

where $n_e = X(z) n_B$. When recombination begins $X(z)$ will decrease.

After a long enough time γ will eventually propagate freely. We call this *decoupling* and is defined by

$$X(z_{\text{dec}}) n_B \sigma_T c \stackrel{!}{=} H(z_{\text{dec}})$$

implying

$$z_{\text{dec}} \sim 1100$$

Note, the γ will of course still scatter with stuff in the Universe.

The CMB is important due observed temperature fluctuations δT defined by

$$\frac{\delta T(\theta, \varphi)}{\langle T \rangle} = \frac{T(\theta, \varphi) - \langle T \rangle}{\langle T \rangle}$$

The largest anisotropy is a dipole anisotropy on the order of 10^{-3} caused by our peculiar motion. We can characterise the dipole by

$$T^* = T \left(1 + \frac{v}{c} \cos \theta \right)$$

with $v \simeq 370 \text{ kms}^{-1}$. We usually remove the dipole after which we are left with fluctuations on the order of 10^{-5} .

We would like to analyse the fluctuations. This is done by expanding in terms of spherical harmonics Y_{lm}

$$\frac{\delta T(\theta, \varphi)}{\langle T \rangle} = \sum_{l=0}^{\infty} \sum_{m=-l}^l a_{lm} Y_{lm}(\theta, \varphi)$$

with

$$a_{lm} = \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin \theta \frac{\delta T}{\langle T \rangle} Y_{lm}$$

Consider the correlation function $C(\alpha)$

$$\begin{aligned} C(\alpha) &= \left\langle \frac{\delta T}{\langle T \rangle}(\hat{u}) \frac{\delta T}{\langle T \rangle}(\hat{u}') \right\rangle_{\hat{u} \cdot \hat{u}' = \cos \alpha} \\ &= \frac{1}{4\pi} \sum_{l=0}^{\infty} (2l+1) C_l P_l(\cos \alpha) \end{aligned}$$

where P_l are the Legendre polynomials and C_l are the multipole moments given by

$$C_l = \frac{1}{2l+1} \sum_{m=-l}^l |a_{lm}|^2$$

The multipole l is related to an angular size α by

$$\alpha = \frac{180^\circ}{l}$$

We work with the *power spectrum* normalised as

$$D_l = \frac{l(l+1)}{2\pi} C_l$$

and

$$\langle \delta T^2 \rangle = D_l \langle T \rangle^2$$

10.5. Ω_i from the CMB

Quantum fluctuations pre-inflation lead to density fluctuations in the early Universe after they are stretched to cosmic scales. These generate potential hills and wells acting on the photon-baryon plasma before recombination. Gravity leads to a pressure in this potential landscape which works to collapse the plasma. We likewise have a radiation pressure which will resist this collapse. The plasma then oscillates in a damped fashion. On potential hills we have rarefaction of density which acts to cool down the plasma, while compressions heats the plasma up.

These oscillations are imprinted on the power spectrum as peaks. The fundamental mode has

$$k_1 = \frac{\pi}{\lambda_{\text{sound}}}$$

with λ_{sound} being the sound horizon. We can imagine the peaks as corresponding to standing waves in the plasma where the temperature anisotropy would be maximal.²⁹

We can immediately determine Ω_k from the power spectrum as the peaks would shift due to γ propagation in a curved Universe being different. The CMB we observe is consistent with $\Omega_k \simeq 0$. This is usually done by using the first peak.

The baryon density Ω_b can be determined by considering the relative size of peaks. Making Ω_b larger would lead to more matter being compressed during infall into potential wells. However, the rebound would be unchanged. This implies compressions or odd peaks are enhanced. When we add mass the oscillations also slow down leading to a shift toward higher l . The matter density Ω_m also affects the peaks as a higher Ω_m would weaken the driving force so peak amplitudes would decrease. The third peak is most sensitive to Ω_m as the first two peaks depend heavily on Ω_b .

²⁹The first peak corresponds to one compression and l gives the scale of λ_{sound} . The second peak corresponds to one compression and one rarefaction, etc.

The damping we observe at large l generally acts like a self-consistency check as it depends on everything.

Also, at small l we have large uncertainties referred to as *cosmic variance* since we only have a single sky we can measure.

Using modern data we can infer

$$\Omega_{\Lambda} \simeq 0.68$$

$$\Omega_c \simeq 0.27$$

$$\Omega_b \simeq 0.05$$

However, there are certain problems with the Λ CDM model presented above such as the *Hubble tension*.

10.6. BBN

Alternative way to determine Ω_b which we do not cover. See any book on cosmology.

11. Dark Matter Candidates

11.1. Requirements for DM

We evidently need a large amount of dark matter, so what is it?

We know:

1. DM is unlikely to consist of baryons. We are quite sure that $\Omega_b \sim 0.05$. This would also be consistent with structure formation.
2. Otherwise:
 1. DM should only interact *weakly* with EM radiation. This is what makes it dark.
 2. DM must have the correct density $\Omega_c \sim 0.25$.
 3. DM must be very light or very heavy, else we would have found them already.
 4. DM must be stable against decay on cosmological scales.

11.2. dMACHOs

There is still the possibility that DM is baryonic and concentrated in *dark massive compact halo objects* or dMACHOs. The belief originally was that DM consisted of BHs, NS, etc. However, we would likely have observed these objects already through gravitational lensing.

11.3. WIMPs

Assuming DM particles are heavy they should interact weakly. We refer to such particles as *weakly interacting massive particles* or WIMPs. We will denote them by χ . One potential candidate is the *neutralino* which is a proposed particle in supersymmetric extensions of the Standard Model. WIMPs should be neutral and if we assume there only is one type of WIMP then it should be its own antiparticle.³⁰

³⁰i.e. a *Majorana particle*.

WIMPs are believed to be produced in the early Universe through particle collisions, in particular through particle-antiparticle collisions. At temperatures $k_B T \gg m_\chi c^2$ particle-antiparticle pairs had enough energy to produce WIMPs at a rate³¹

$$\Gamma_\chi = \langle \sigma v \rangle n_\chi$$

These production processes would have been in equilibrium with WIMP annihilation. When the Universe expanded the temperature decreased and the number of particles able to produce WIMPs would drop exponentially leading to an equilibrium density

$$n_\chi \sim \left(\frac{m_\chi T}{2\pi} \right)^{3/2} \exp \left[-\frac{m_\chi c^2}{k_B T} \right]$$

Also, n_χ would decrease due to the expansion of the Universe. When

$$\Gamma_\chi \sim H$$

WIMP production decouples. After this point n_χ decreases as a^{-3} . We can show the decoupling temperature T_χ is given by

$$k_B T_\chi \sim \frac{m_\chi c^2}{x}$$

with $20 \lesssim x \lesssim 50$ for $10 \text{ GeV} \lesssim m_\chi c^2 \lesssim 10 \text{ TeV}$ and

$$\frac{\Omega_\chi}{0.2} \sim \left(\frac{m_\chi c^2 / k_B T}{20} \right) \left(\frac{3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}}{\langle \sigma v \rangle} \right)$$

Then the *relic density* decreases as $\langle \sigma v \rangle$ increases. This is the case since the WIMPs are able to stay in thermodynamic equilibrium for longer, which suppresses their density by the Boltzmann factor in n_χ .

Assuming $x \sim 20$ and $v \sim \frac{1}{3}c$ we require $\sigma \sim 3 \text{ pb}$ for the proper Ω_χ . Taking the interaction to be on the order of the weak interaction $g_\chi \sim g_W$ then

$$\sigma \sim \frac{g_W^4}{m_\chi^2}$$

implying

³¹We assume DM is cold. This is consistent with observed LSS.

$$m_\chi \sim 100 \text{ GeV}$$

for $\sigma \sim 3\text{pb}$. This is called the *WIMP miracle*.

11.4. Detection of WIMPs

The detection candidates for many DM candidates are generally model dependent. We necessarily also carry a prejudice with us when considering candidates since we can only probe certain things. We always need some *signature* unique to a given model!

One can imagine many different ways to try and observe DM using astroparticle observables.³² We can generally consider three different search strategies. We can attempt indirect detection by considering a process of the form

$$\underbrace{\chi + \chi}_{\text{dark sector}} \xrightarrow{\text{model dependent process}} \underbrace{f + \bar{f}}_{\text{Standard Model}}$$

with χ being a DM candidate and f being some known particle in the Standard Model. We could then possibly observe f and $\bar{f}$ and try to check predictions from some model. Likewise, we could attempt to go the other way by a process of the form

$$f + \bar{f} \rightarrow \chi + \chi$$

using colliders. Lastly, we could possibly conceive a detector being able to directly detect χ .

11.5. Indirect searches for DM

Consider the case of WIMP DM annihilation. We could have annihilation of the form

$$\chi + \chi \xrightarrow{\text{model dependent}} \begin{cases} W^-/Z/q \\ W^+/Z/\bar{q} \end{cases} \rightarrow \text{decays} \begin{cases} \pi^0 \\ \pi^+ \\ \pi^- \end{cases} \xrightarrow{\text{further decays}} \begin{cases} \gamma\text{-rays} \\ \nu \\ p/\bar{p}, d/\bar{d}, \text{ anti-matter} \end{cases}$$

or

$$\chi + \chi \rightarrow \underbrace{l + \bar{l}}_{\text{lepton pairs}} \xrightarrow{\text{Bremsstrahlung, Synchrotron, Compton}} \gamma$$

or

$$\chi + \chi \rightarrow \gamma$$

³²e.g. γ - and cosmic rays.

The expected γ flux from WIMP annihilation can be written as

$$\frac{d\Phi_\gamma}{dE_\gamma d\Omega}(E_\gamma, \Psi) = \overbrace{\frac{1}{4\pi} \int_{\text{l.o.s}} \rho_\chi^2(\mathbf{r}) d\mathcal{L}(\Psi)}^{\text{amount of DM}} \underbrace{\left[\frac{\langle\sigma v\rangle_{\text{ann}}}{2S_\chi m_\chi^2} \sum_f \frac{dN_\gamma^f}{dE_\gamma} B_f \right]}_{\text{amount of } \gamma}$$

where Ψ is the direction of observation. $d\mathcal{L}(\Psi)$ is the line of sight. $\rho_\chi(\mathbf{r})$ is the DM density at the point $\mathbf{r}$, which we square because we are talking about annihilation. $\langle\sigma v\rangle_{\text{ann}}$ is the velocity averaged annihilation cross section. S_χ is the symmetry factor. dN_γ^f/dE_γ is the γ -ray spectrum for each annihilation channel f with branching ratio B_f . We can write

$$\frac{dJ}{d\Omega} = \frac{1}{4\pi} \int_{\text{l.o.s}} \rho_\chi^2(\mathbf{r}) d\mathcal{L}(\Psi) \sim \text{astrophysics}$$

Then

$$\frac{d\Phi_\gamma}{dE_\gamma}(E_\gamma, \Delta\Omega) = \underbrace{\frac{1}{4\pi} \int_{\Delta\Omega} d\Omega \int_{\text{l.o.s}} \rho_\chi^2(\mathbf{r}) d\mathcal{L}(\Psi)}_{J \sim \text{astrophysics}} \overbrace{\left[\frac{\langle\sigma v\rangle_{\text{ann}}}{2S_\chi m_\chi^2} \sum_f \frac{dN_\gamma^f}{dE_\gamma} B_f \right]}^{\text{particle physics}}$$

Assuming DM has uniform density ρ_χ is annihilating in a spherical galaxy of radius r at distance d we find

$$J \simeq \frac{4\pi r^3 \rho_\chi^2}{3d^2}$$

Then nice search targets have high ρ_χ are close (small d), large (big V) and have well understood backgrounds.

An example people use are dwarf spheroidal galaxies in the DM halo of the Milky Way. Assuming steady state and spherical symmetry one can approximate the DM density through the 2nd Jeans equation which then provides an estimate of the J factor.³³ The dSphs also have high mass to light ratios and little background contamination, thereby making them excellent search targets.

³³using measured velocities of stars.

The idea is then to pick our favourite dSph and compute the corresponding J factor.

We can model the emissions as

$$\frac{d\Phi_\gamma}{dE_\gamma}(\ell, b) = \underbrace{\frac{d\Phi_{\gamma, \text{diffuse}}}{dE_\gamma} + \frac{d\Phi_{\gamma, \text{PS}}}{dE_\gamma}}_{\text{background}} + \underbrace{\frac{d\Phi_{\gamma, \text{DM}}}{dE_\gamma}}_{\text{DM signal}}$$

We can then compare with γ -ray data using the maximum likelihood method and see if a model including DM is favorable. One uses the Poisson likelihood

$$\mathcal{L}(m_\chi, \langle\sigma v\rangle, \theta \mid D_{ij}) = \prod_{i,j} \frac{e^{-\mu_{ij}} \mu_{ij}^{N_{ij}}}{N_{ij}!}$$

where N_{ij} are the number of observed γ -ray counts and $\mu_{ij} \equiv \mu_{ij}(m_\chi, \langle\sigma v\rangle, \theta)$ are the expected number of counts. Here, θ represents all nuisance parameters which are not of interest. We can compute μ_{ij} by various methods. Then we can do a likelihood ratio test

$$\text{TS} = -2 \sum_{i,j} \ln \left[\frac{\mathcal{L}(m_\chi, \langle\sigma v\rangle \stackrel{!}{=} 0, \hat{\theta} \mid D_{ij})}{\mathcal{L}(m_\chi, \widehat{\langle\sigma v\rangle}, \hat{\theta} \mid D_{ij})} \right]$$

where $\hat{\square}$ represents parameters optimised in a fit to data with fixed m_χ . When TS is large their is a preference for including DM.

When people have done the above they found no significant detection of a DM signal. However, the data we have can be used to constrain $\langle\sigma v\rangle$. This is simply done by another hypothesis test comparing a fixed $\langle\sigma v\rangle$ to the global best fit value. These limits obviously become stronger when more dSphs are included.

Aside from using dSphs many other astrophysical sources can obviously be used, however, this is very difficult since one needs to be able to confidently model the background. As an example there has been observed an excess signal from the galactic center. This excess could be due to DM annihilation, but it could also be from various other things. Many of these alternative explanations will probably be ruled out or confirmed

eventually as we get better models and detectors.³⁴

11.6. Direct searches for DM

Consider the elastic scattering of WIMPs on nucleons

$$\chi + N \rightarrow \chi + N$$

We use the lab frame wherein N is stationary. Then

$$\mathbf{p}_\chi = m_\chi \mathbf{v}; \quad \mathbf{p}_N = 0$$

After scattering we have

$$\mathbf{q}' = m_\chi \mathbf{v}'; \quad |\mathbf{q}| = \sqrt{2m_N E_{nr}}$$

with scattering angles θ' and θ .

The velocity of collisionless DM particles should follow the Maxwell-Boltzmann distribution in a non-rotating halo

$$f(\mathbf{v}) = N|\mathbf{v}|^2 \exp\left[-\frac{3|\mathbf{v}|^2}{2\sigma^2}\right]$$

with

$$\sigma(R) = \sqrt{\frac{3}{2}} v_c(R)$$

$$v_{c(R_\odot)} = 220 \text{ kms}^{-1}$$

The distribution is truncated at $v_{\text{esc}} = \sqrt{2\phi} \simeq 544 \text{ kms}^{-1}$ with ϕ being the Milky Way gravitational potential. Note, since the Earth goes around the Sun v_c varies annually.

The WIMP scattering rate is given by

$$\frac{dR}{dE_{nr}} = \frac{\rho_\chi}{m_\chi} \frac{M}{m_N} \int_{v_{\min}}^{v_{\text{esc}}} v f(v) \frac{d\sigma}{dE_{nr}} dv \propto \exp\left[-\frac{4\mu E_{nr}}{E_0}\right]$$

With M being the target detector mass. $v_{\min}$ is given by

$$v_{\min} = \sqrt{\frac{E_{nr} m_N}{2} \frac{1}{\mu^2}}$$

³⁴e.g. emission from millisecond pulsars is favored.

μ is the reduced mass and E_0 is the most probable interaction energy. The number of observed recoils in a time T is given by

$$N_{\text{obs}} = T \int_{E_{\text{min}}}^{E_{\text{max}}} dE_{nr} \varepsilon(E_{nr}) \frac{dR}{dE_{nr}}$$

with ε being the efficiency of detecting recoil at E_{nr} and

$$E_{\text{max}} = \frac{2\mu v_{\text{esc}}}{m_N}$$

$$E_{\text{min}} \sim \text{set by detector}$$

The WIMP scattering cross section is given by

$$\frac{d\sigma}{dE_{nr}} = \frac{m_N}{2v^2\mu^2} \left[\underbrace{\sigma_{\text{SI}} F_{\text{SI}}^2(E_{nr})}_{\text{form factor}} + \sigma_{\text{SD}} F_{\text{SD}}^2(E_{nr}) \right]$$

with $\square_{\text{SI}}$ being the spin independent part arising from scalar or vector interactions and $\square_{\text{SD}}$ being the spin dependent part arising from axial vector interactions

$$\mathcal{L}_S \sim \bar{\chi}\chi\bar{q}q$$

$$\mathcal{L}_V \sim \bar{\chi}\gamma^\mu\chi\bar{q}\gamma_\mu q$$

$$\mathcal{L}_A \sim \bar{\chi}\gamma^\mu\gamma_5\chi\bar{q}\gamma_\mu\gamma_5 q$$

We can write

$$\sigma_{\text{SI}} = \frac{\sigma_n \mu^2 A^2}{\mu_n^2}$$

with σ_n being the WIMP nucleon cross section, μ_n is the reduced mass of the WIMP nucleon system, and A is the atomic number. The σ_{SD} is more complicated and depends on many things.

We can write the energy loss of WIMPs in detectors as

$$\left(\frac{dE}{dx}\right)_{\text{tot}} = \underbrace{\left(\frac{dE}{dx}\right)_{\text{elec}}}_{\text{WIMP electron interactions excitation or ionisation}} + \underbrace{\left(\frac{dE}{dx}\right)_{\text{nucl}}}_{\text{WIMP nucleon scattering heat} \rightarrow \text{phonons}}$$

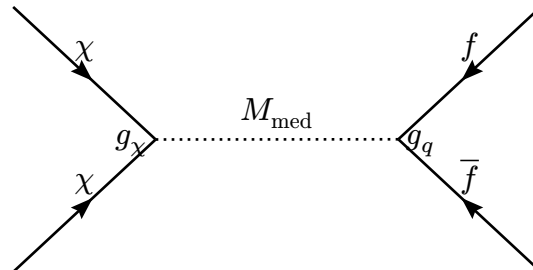
This leads to a breadth of different detectors trying to detect side effects of a recoiling nucleus:

1. Anorganic crystals: E_{kin} of recoiling nucleus
 1. in semiconducting crystals creates electron-hole pairs.
 2. in scintillating crystals excites and ionises atoms in the crystal lattice. This leads to scintillation light.
2. Cryogenic detectors: E_{kin} of recoiling nucleus
 1. leads to heating of crystals and the creation of phonons.
3. Noble liquids in TPCs: E_{kin} of recoiling nucleus
 1. leads to heat.
 2. excited states X^* detected through scintillation.
 3. ionised states detected through scintillation (S1). $\mathbf{E}$ -field removes ionised e^- leading to secondary scintillation signal (S2) in gaseous phase.
4. Bubble chambers: E_{kin} of recoiling nucleus
 1. creates bubbles in a superheated liquid.

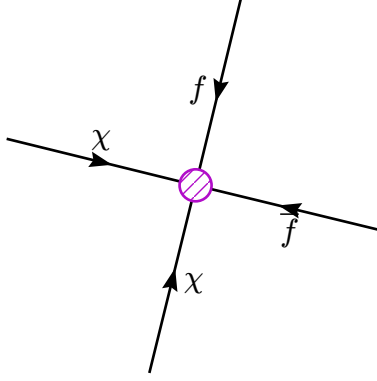
All of the above are heavily affected by backgrounds which must be reduced. Many detectors have managed to heavily constrain the scattering cross section.

11.7. Complementarity

Comparing direct with indirect searches is usually not possible. We therefore use simplified models to compare with collider searches



We consider EFT to compare direct and indirect searches. Assuming $E_{\text{c.o.m}} \ll M_{\text{med}}$ we have



with the EFT scale $M_\star$ given by

$$M_\star^2 \sim \frac{M_{\text{med}}}{g_\chi g_q}$$

Different interactions suffer from different suppressions so one typically combines search strategies. As an example

$$\text{scalar interactions} \sim \frac{m_q}{M_\star^3} (\bar{\chi}\chi)(\bar{q}q) \sim \underbrace{v^2}_{\text{indirect suppression}} \quad \text{and} \quad \overbrace{1}^{\text{direct suppression}}$$

CTA upper limits on the cross section can be converted to lower limits on the EFT scale $M_\star$.